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Liao et al.

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(54) **INTERCONNECTION STRUCTURE AND METHODS OF FORMING THE SAME**

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H01L 21/768 (2006.01)
H01L 23/532 (2006.01)

(52) **U.S. Cl.**
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(58) **Field of Classification Search**

CPC H01L 23/528; H01L 21/76802; H01L 23/53238; H01L 23/53266; H01L 21/76841; H01L 21/7685; H01L 21/76834; H01L 21/76885; H01L 23/4824; H01L 21/0332; H01L 21/76895; H01L 23/4827

See application file for complete search history.

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Primary Examiner — Thanh T Nguyen

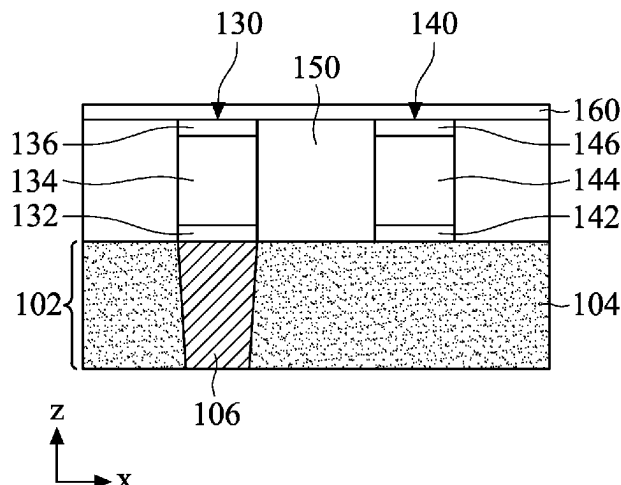
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(57) **ABSTRACT**

An interconnect structure includes a dielectric layer, a first conductive feature, a second conductive feature, a third conductive feature, and a dielectric fill. The first conductive feature is disposed in the dielectric layer. The second conductive feature is disposed over the first conductive feature. The second conductive feature includes a first conductive layer disposed over the first conductive feature, a second conductive layer disposed on the first conductive layer, and a third conductive layer disposed on the second conductive layer. The first conductive layer, the second conductive layer and the third conductive layer have substantially the same width. The third conductive feature is disposed over the dielectric layer. The dielectric fill is disposed over the dielectric layer between the second conductive feature and the third conductive feature.

18 Claims, 13 Drawing Sheets

101



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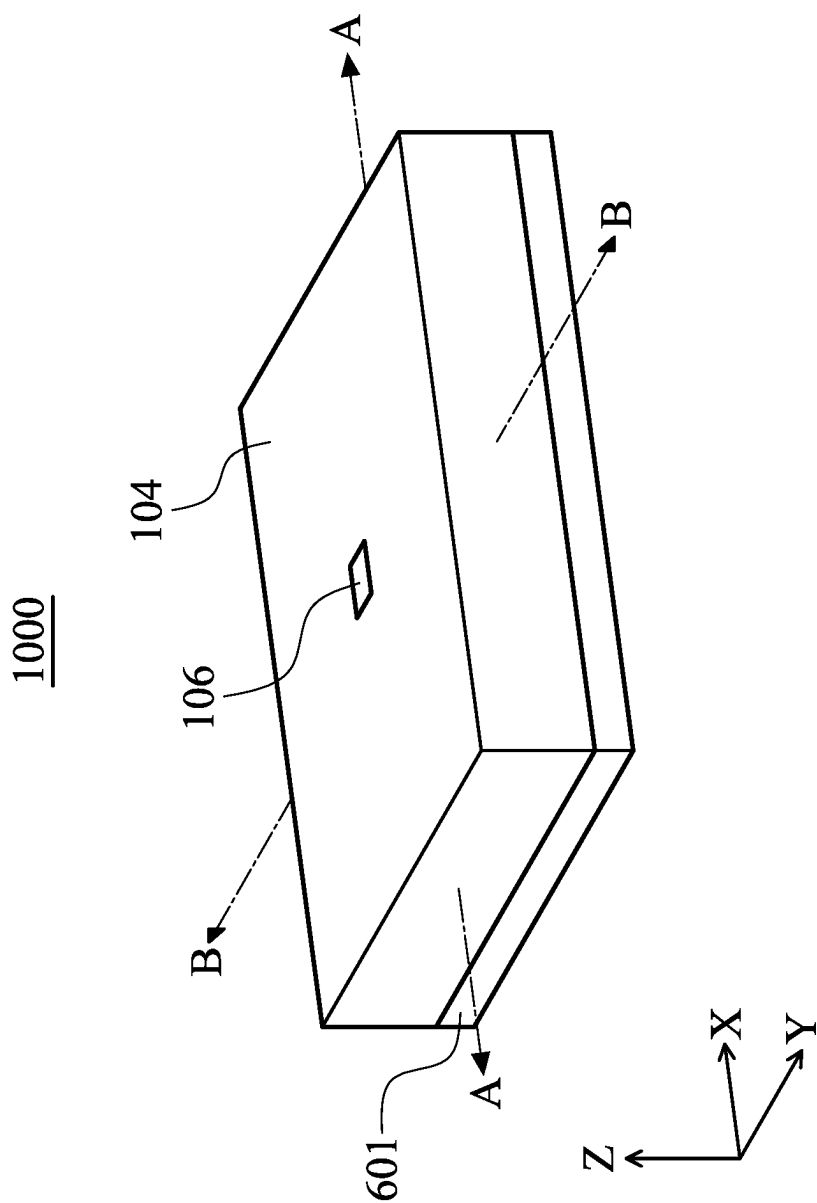


FIG. 1

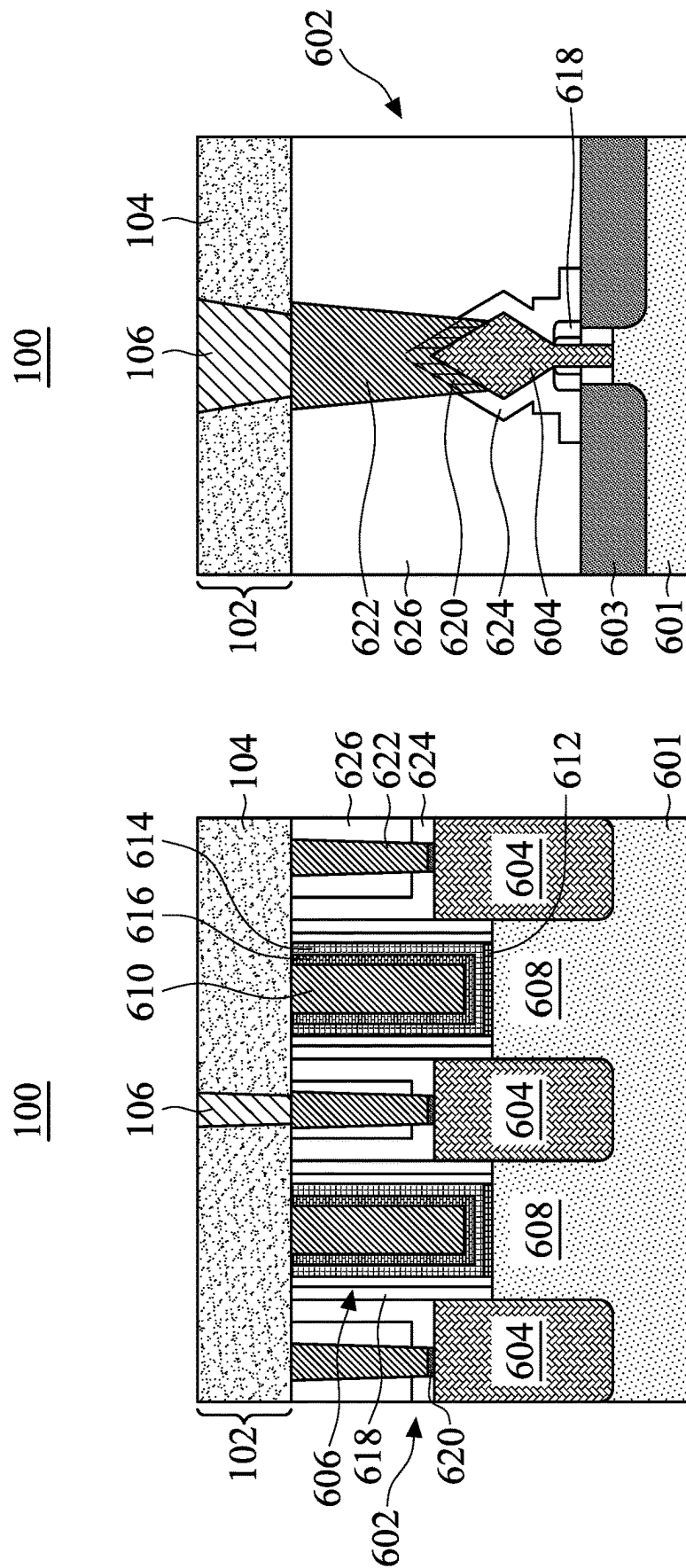


FIG. 2A

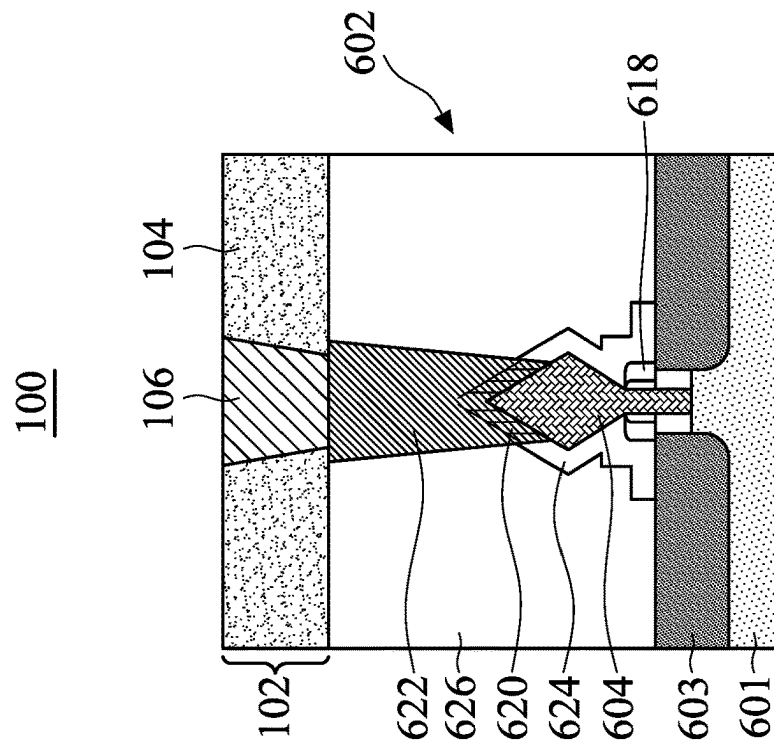


FIG. 2B

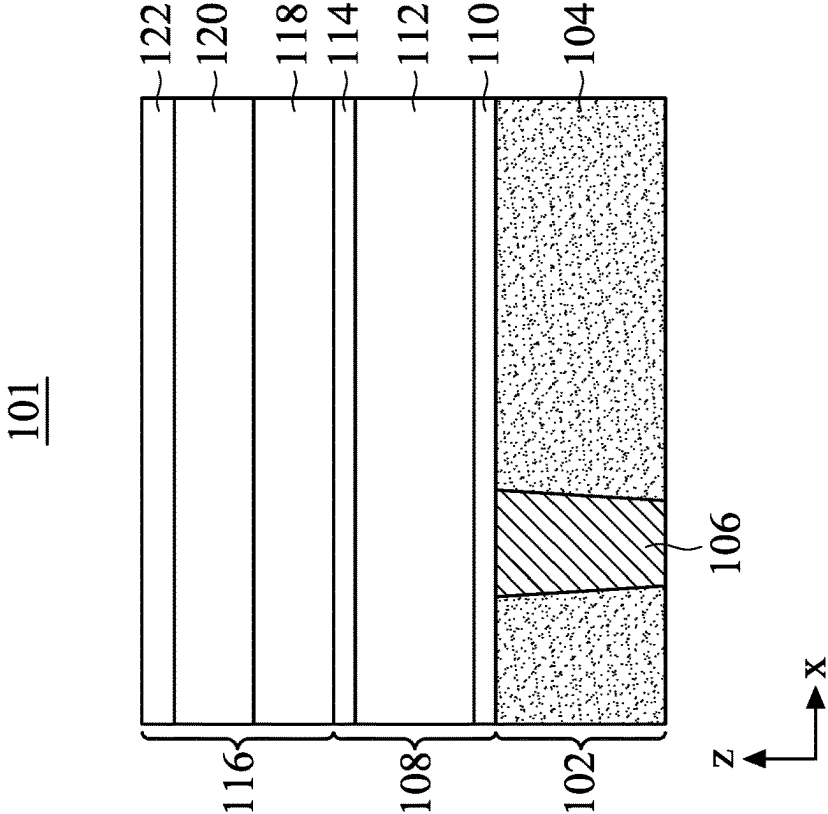


FIG. 3B

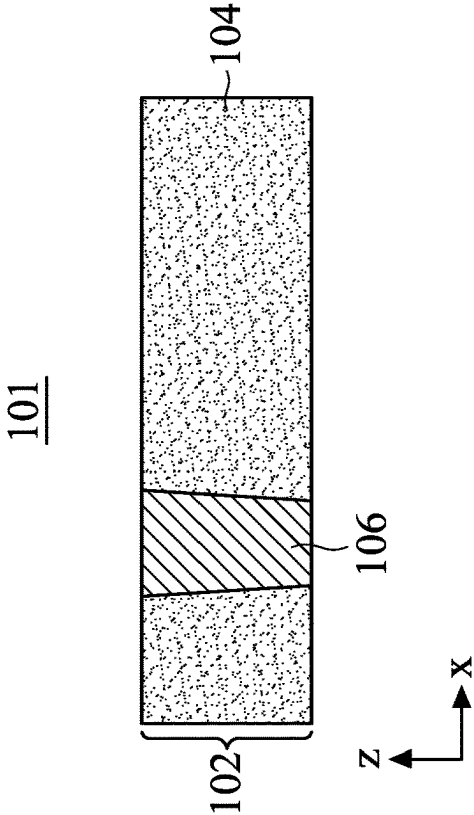
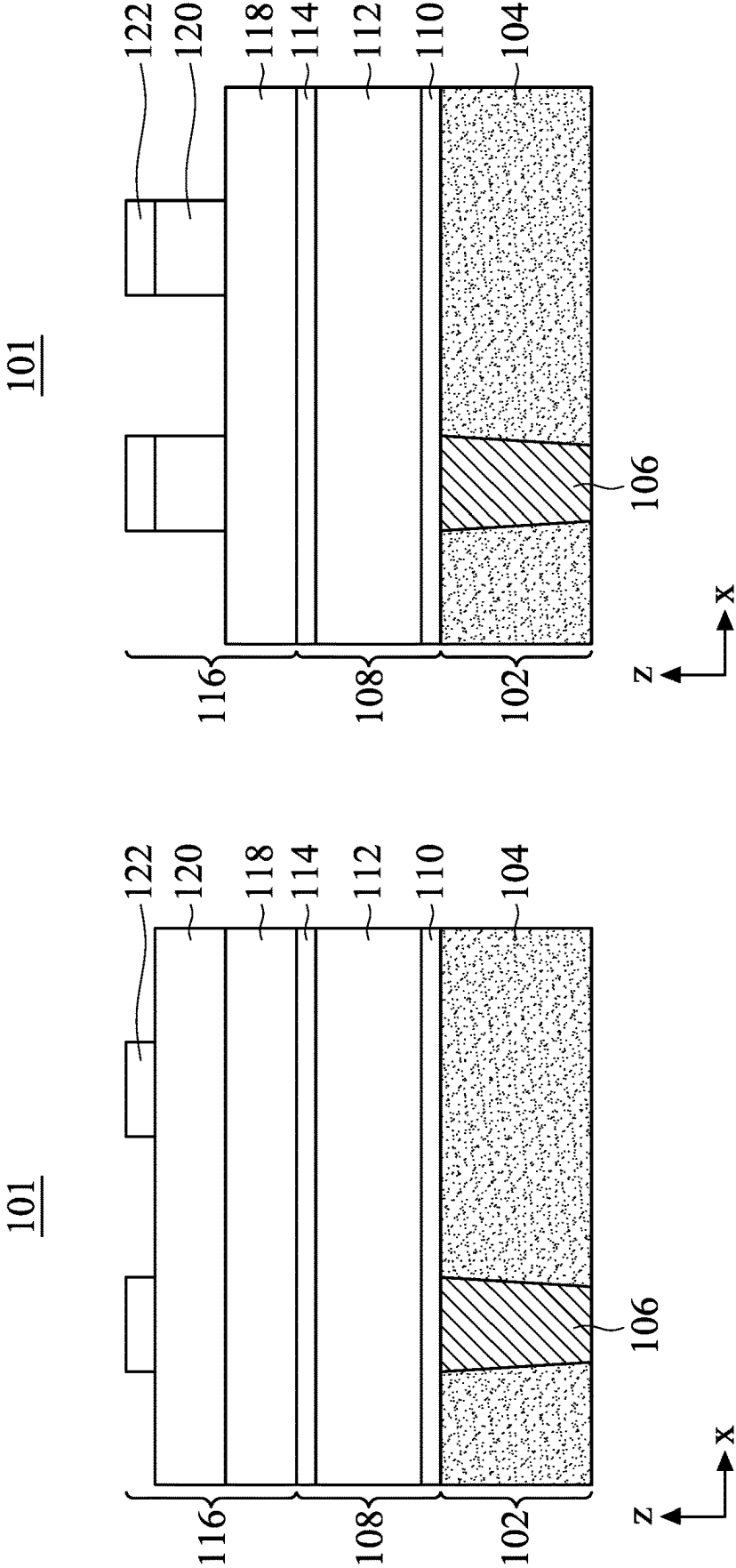
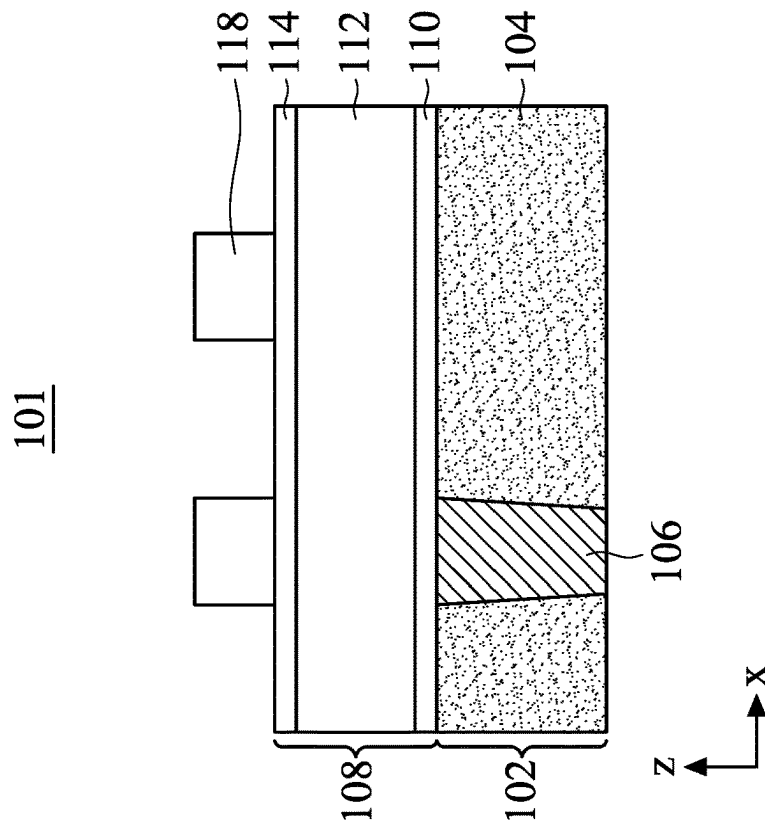
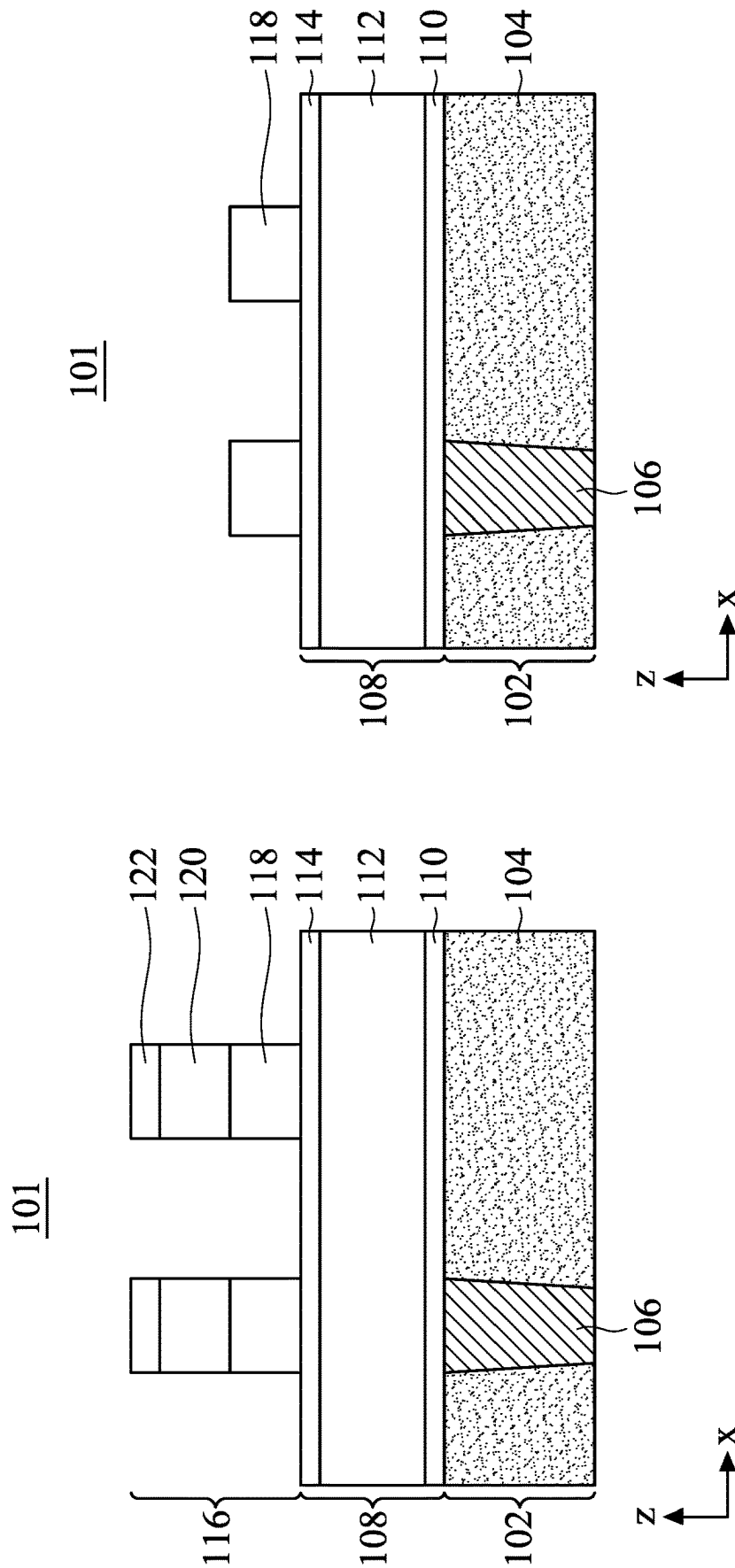
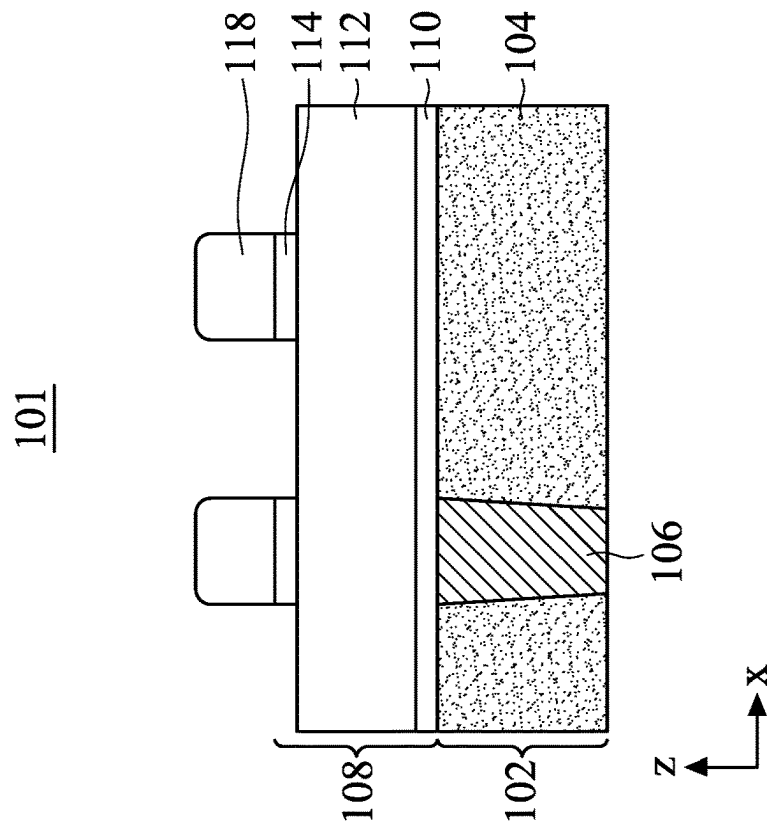
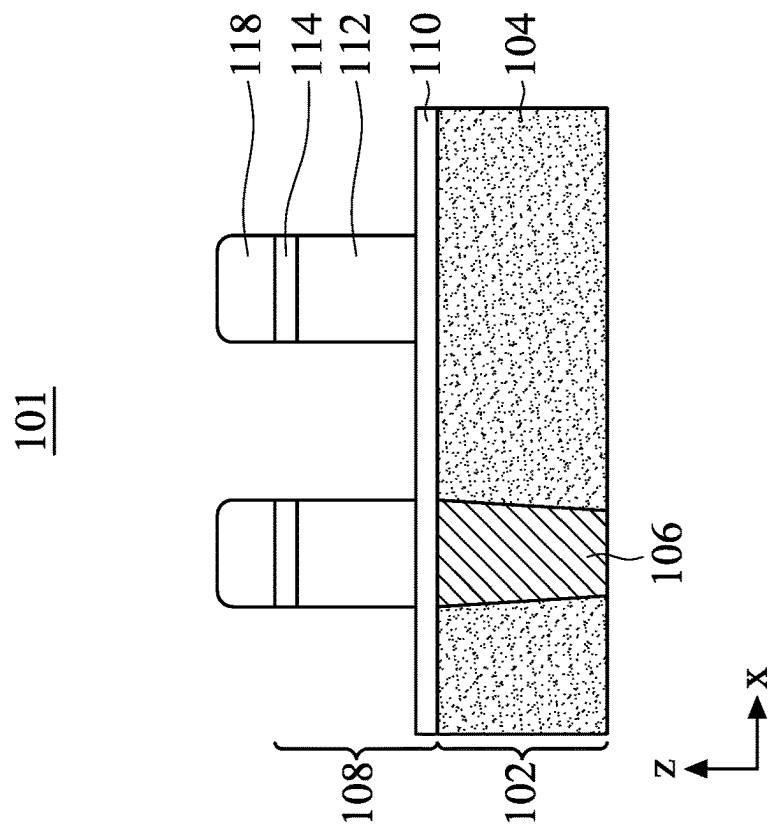


FIG. 3A







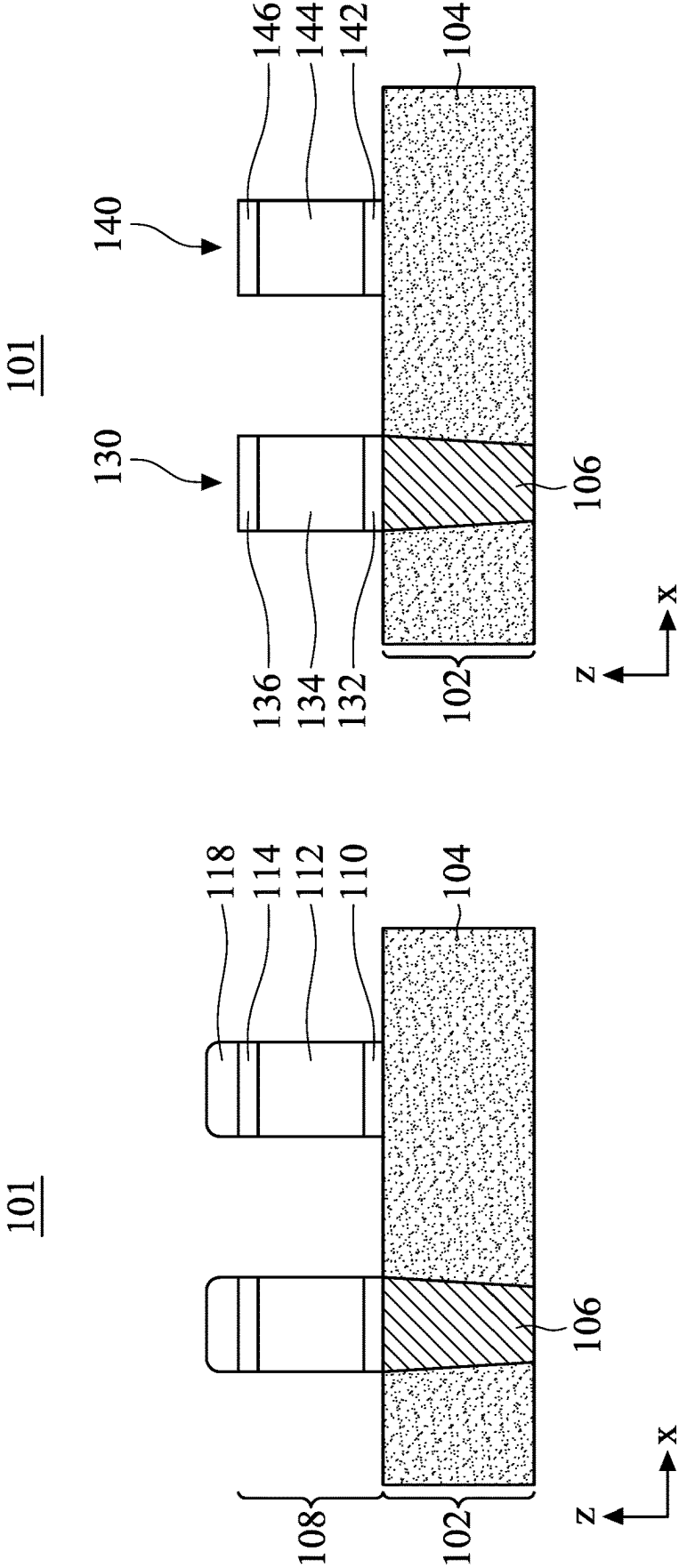


FIG. 3I

FIG. 3J

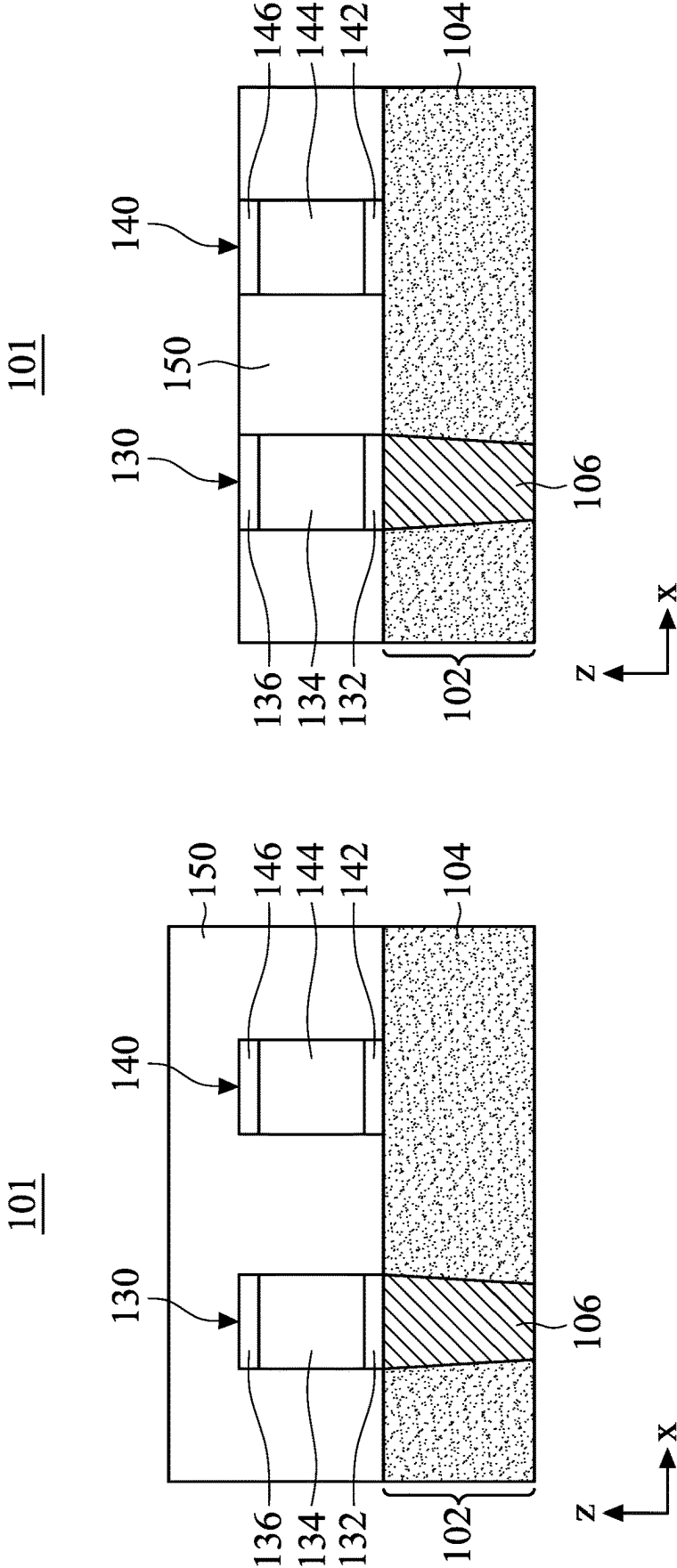


FIG. 3L

FIG. 3K

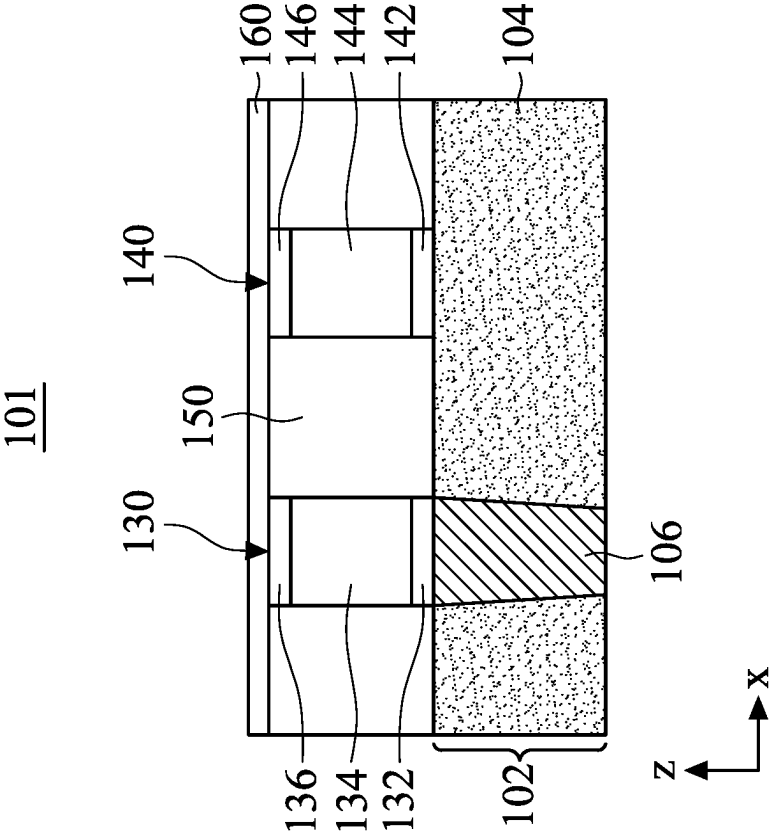


FIG. 3M

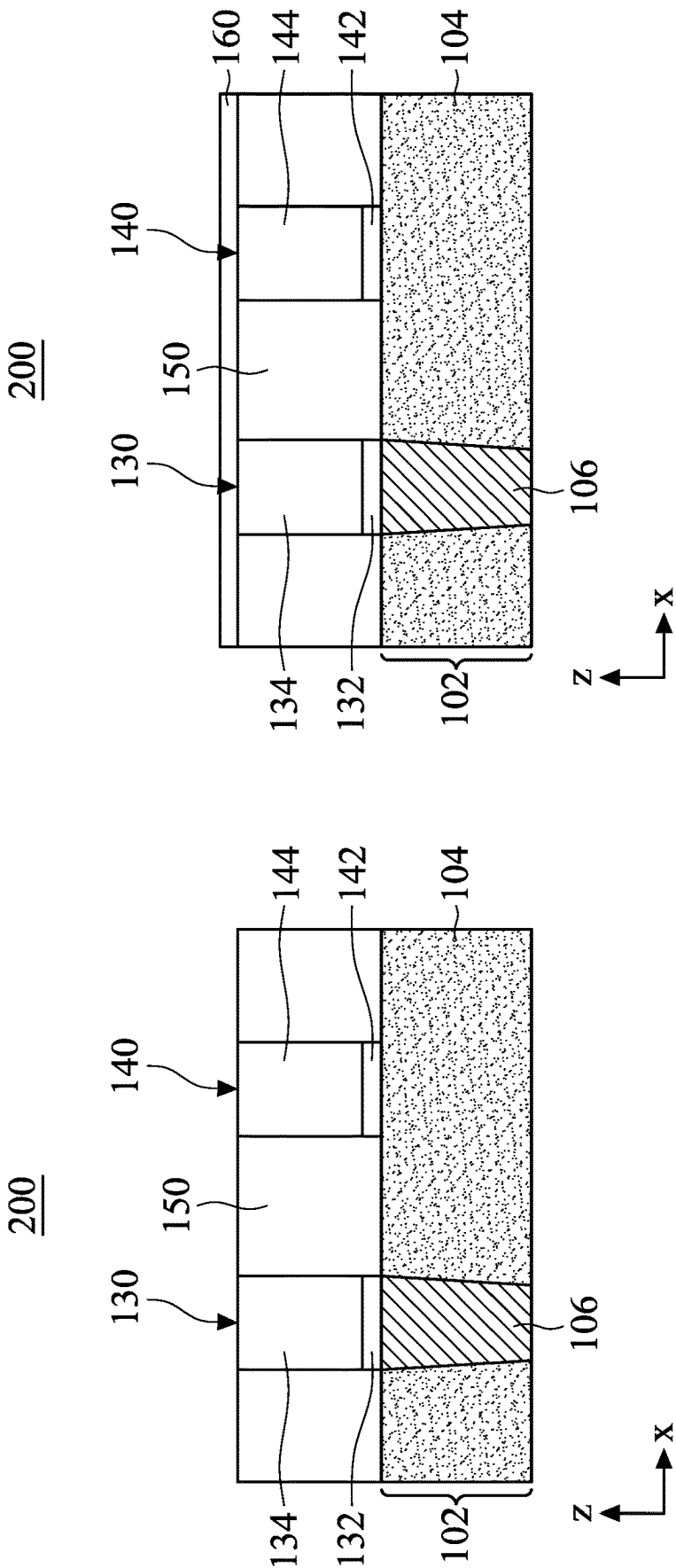
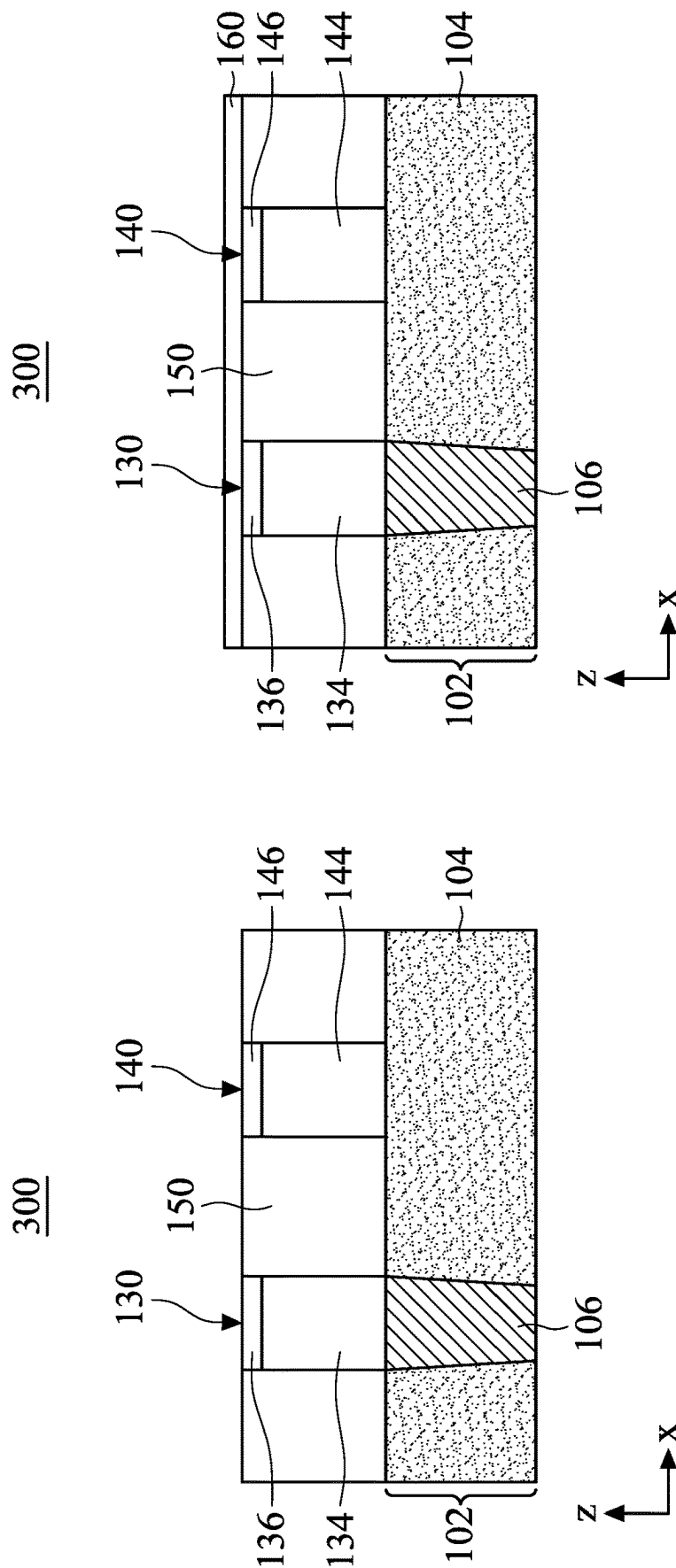
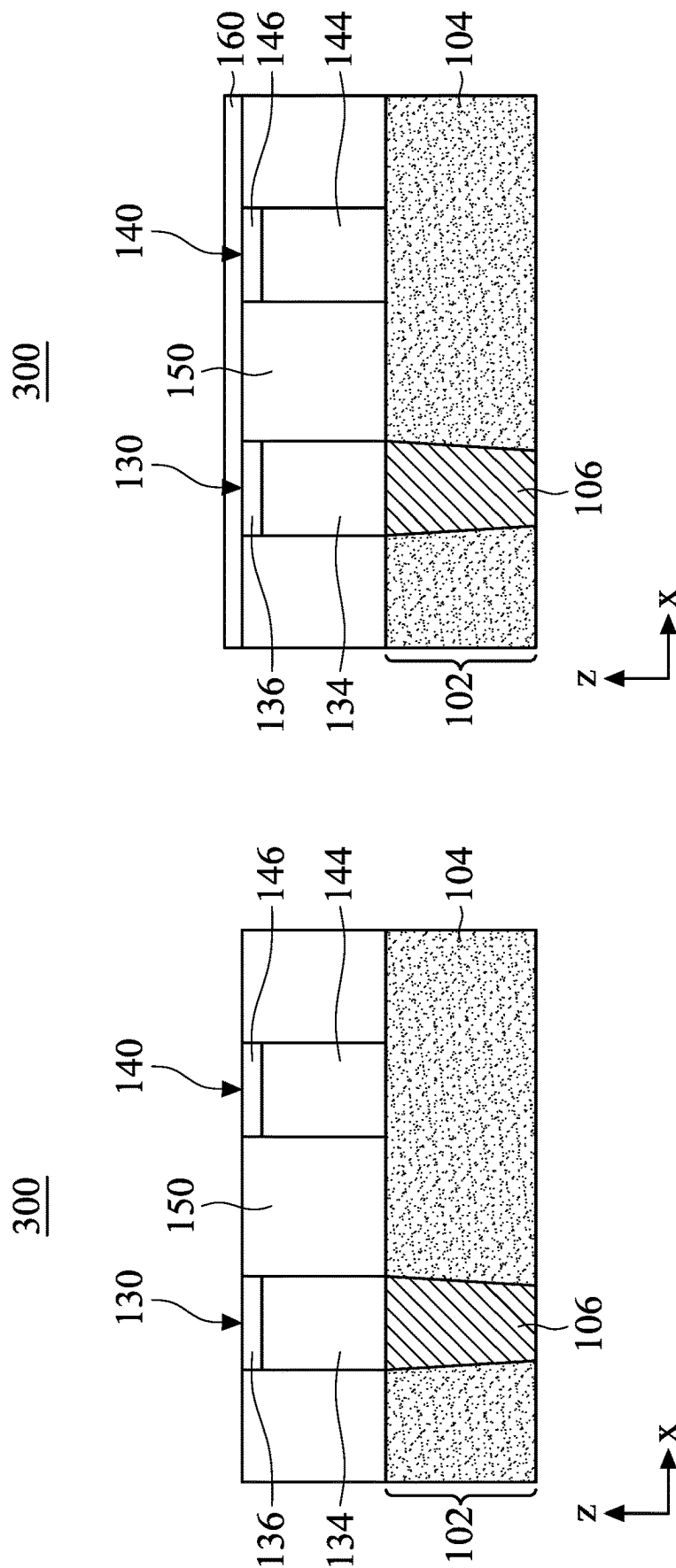


FIG. 4B

FIG. 4A



400

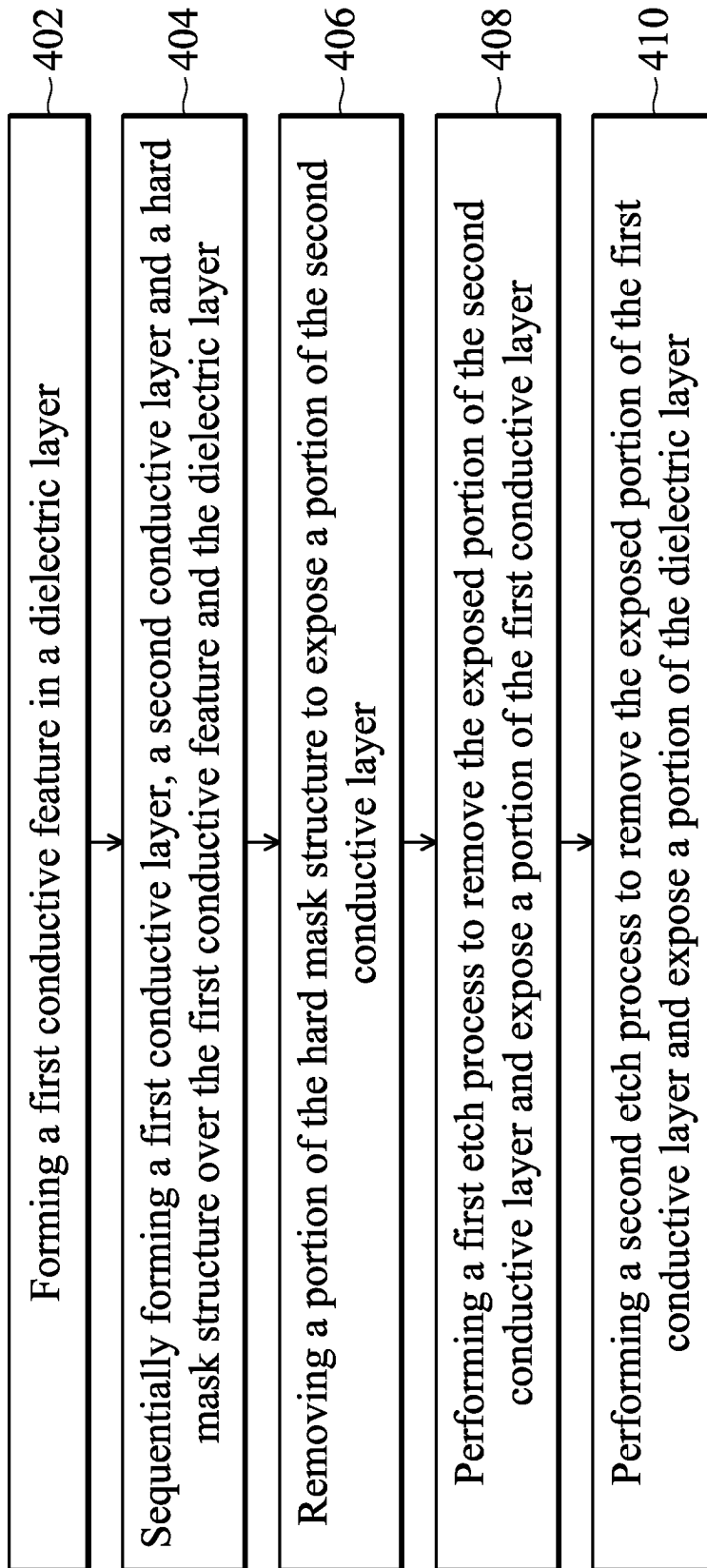


FIG. 6

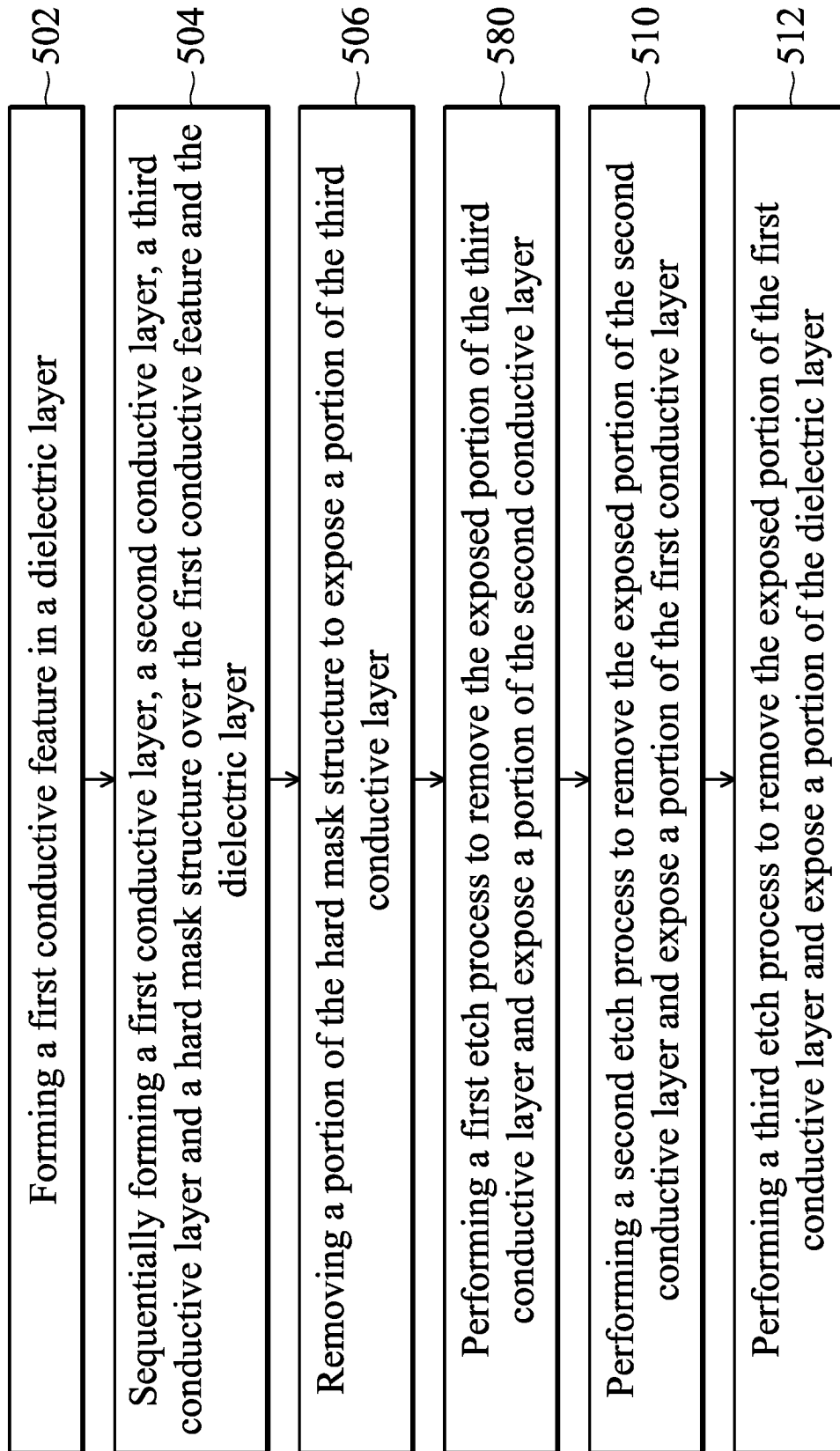
500

FIG. 7

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INTERCONNECTION STRUCTURE AND METHODS OF FORMING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional application of U.S. patent application Ser. No. 17/313,217 filed May 6, 2021, which is incorporated by reference in its entirety.

BACKGROUND

Device scaling, which has driven the semiconductor technology for decades, increases not only transistor density, but also that of the metal interconnects. As the semiconductor industry introduces new generations of integrated circuits (ICs) with higher performance and more functions, the density of components forming the ICs has increased, while the dimensions, sizes, and spacing between components or elements have decreased.

The main purpose of continuous scaling of the device dimensions is to improve the performance of the semiconductor microprocessors and to pack more devices in the same area. However, as the technology node is advanced, the distances between metal contacts or metal lines become much shorter and the difficulty of the etching process to form the metal lines has also increased.

Therefore, there is a need in the art to provide improved structures and methods that can address the issues mentioned above.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 is a perspective view of one of the various stages of manufacturing a semiconductor device structure, in accordance with some embodiments.

FIGS. 2A-2B are cross-sectional side views of various stages of manufacturing an interconnect structure, in accordance with some embodiments.

FIGS. 3A-3M are cross-sectional side views of various stages of manufacturing an interconnect structure, in accordance with some embodiments.

FIGS. 4A-4B are cross-sectional side views of various stages of manufacturing another interconnect structure, in accordance with some embodiments.

FIGS. 5A-5B are cross-sectional side views of various stages of manufacturing a further interconnect structure, in accordance with some embodiments.

FIG. 6 is a flow chart of a method of manufacturing an interconnect structure in accordance with some embodiments.

FIG. 7 is another flow chart of a method of manufacturing an interconnect structure in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of

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components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “on,” “top,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

FIG. 1 is a perspective view of one of the various stages of manufacturing a semiconductor device structure **100**, in accordance with some embodiments. As shown in FIG. 1, the semiconductor device structure **100** includes a substrate **601** having at least a plurality of conductive features formed thereover (one conductive feature **106** is shown in FIG. 1). The conductive feature **106** is formed in a dielectric material **104**. One or more devices (not shown), such as transistors, diodes, imaging sensors, resistors, capacitors, inductors, memory cells, a combination thereof, and/or other suitable devices, may be formed between the substrate **601** and the conductive feature **106**.

FIGS. 2A-2B, 3A-3M, 4A-4B and 5A-5B show exemplary sequential processes for manufacturing a semiconductor device structure **100**, in accordance with some embodiments. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. 2A-2B, 3A-3M, 4A-4B and 5A-5B, and some of the operations described below can be replaced or eliminated, for additional embodiments of the process. The order of the operations/processes may be interchangeable.

FIG. 2A is a cross-sectional side view of the semiconductor device structure **100** taken along line A-A of FIG. 1, and FIG. 2B is a cross-sectional side view of the semiconductor device structure **100** taken along line B-B of FIG. 1. The line A-A of FIG. 1 extends along a direction that is substantially perpendicular to the longitudinal direction of a gate stack **606**, and the line B-B of FIG. 1 extends along the longitudinal direction of the gate stack **606**. As shown in FIGS. 2A and 2B, the semiconductor device structure **100** includes the substrate **601**, one or more devices **602** formed on the substrate **601**, the dielectric material **104** formed over the devices **602**, and the conductive features **106** formed in the dielectric material **104**. The substrate **601** may be a semiconductor substrate. In some embodiments, the substrate **601** includes a single crystalline semiconductor layer on at least the surface of the substrate **601**. The substrate **601** may include a single crystalline semiconductor material such as, but not limited to silicon (Si), germanium (Ge), silicon germanium (SiGe), gallium arsenide (GaAs), indium antimonide (InSb), gallium phosphide (GaP), gallium antimonide (GaSb), indium aluminum arsenide (InAlAs),

indium gallium arsenide (InGaAs), gallium antimony phosphide (GaSbP), gallium arsenic antimonide (GaAsSb), and indium phosphide (InP). For example, the substrate **601** is made of Si. In some embodiments, the substrate **601** is a silicon-on-insulator (SOI) substrate, which includes an insulating layer (not shown) disposed between two silicon layers. In one aspect, the insulating layer is an oxygen-containing material, such as an oxide.

The substrate **601** may include one or more buffer layers (not shown) on the surface of the substrate **601**. The buffer layers can serve to gradually change the lattice constant from that of the substrate to that of the source/drain regions. The buffer layers may be formed from epitaxially grown crystalline semiconductor materials such as, but not limited to Si, Ge, germanium tin (GeSn), SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, GaN, GaP, and InP. In one embodiment, the substrate **601** includes SiGe buffer layers epitaxially grown on the silicon substrate **601**. The germanium concentration of the SiGe buffer layers may increase from 30 atomic percent germanium for the bottom-most buffer layer to 70 atomic percent germanium for the top-most buffer layer.

The substrate **601** may include various regions that have been suitably doped with impurities (e.g., p-type or n-type impurities). The dopants are, for example phosphorus for an n-type fin field effect transistor (FinFET) and boron for a p-type FinFET.

As described above, the devices **602** may be any suitable devices, such as transistors, diodes, imaging sensors, resistors, capacitors, inductors, memory cells, or a combination thereof. In some embodiments, the devices **602** are transistors, such as planar field effect transistors (FETs), FinFETs, nanostructure transistors, or other suitable transistors. The nanostructure transistors may include nanosheet transistors, nanowire transistors, gate-all-around (GAA) transistors, multi-bridge channel (MBC) transistors, or any transistors having the gate electrode surrounding the channels. An example of the device **602** formed between the substrate **601** and the conductive features **106** is a FinFET, which is shown in FIGS. 2A and 2B. The device **602** includes source/drain (S/D) regions **604** and gate stacks **606**. Each gate stack **606** may be disposed between S/D regions **604** serving as source regions and S/D regions **604** serving as drain regions. For example, each gate stack **606** may extend along the Y-axis between a plurality of S/D regions **604** serving as source regions and a plurality of S/D regions **604** serving as drain regions. As shown in FIG. 2A, two gate stacks **606** are formed on the substrate **601**. In some embodiments, more than two gate stacks **606** are formed on the substrate **601**. Channel regions **608** are formed between S/D regions **604** serving as source regions and S/D regions **604** serving as drain regions.

The S/D regions **604** may include a semiconductor material, such as Si or Ge, a III-V compound semiconductor, a II-VI compound semiconductor, or other suitable semiconductor material. Exemplary S/D region **604** may include, but are not limited to, Ge, SiGe, GaAs, AlGaAs, GaAsP, SiP, InAs, AlAs, InP, GaN, InGaAs, InAlAs, GaSb, AlP, GaP, and the like. The S/D regions **604** may include p-type dopants, such as boron; n-type dopants, such as phosphorus or arsenic; and/or other suitable dopants including combinations thereof. The S/D regions **604** may be formed by an epitaxial growth method using CVD, atomic layer deposition (ALD) or molecular beam epitaxy (MBE). The channel regions **608** may include one or more semiconductor materials, such as Si, Ge, GeSn, SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, GaN, GaP, or InP. In

some embodiments, the channel regions **608** include the same semiconductor material as the substrate **601**. In some embodiments, the devices **602** are FinFETs, and the channel regions **608** are a plurality of fins disposed below the gate stacks **606**. In some embodiments, the devices **602** are nanosheet transistors, and the channel regions **608** are surrounded by the gate stacks **606**.

Each gate stack **606** includes a gate electrode layer **610** disposed over the channel region **608** (or surrounding the channel region **608** for nanostructure transistors). The gate electrode layer **610** may be a metal-containing material such as tungsten, cobalt, aluminum, ruthenium, copper, multilayers thereof, or the like, and can be deposited by ALD, plasma enhanced chemical vapor deposition (PECVD), MBD, physical vapor deposition (PVD), or any suitable deposition technique. Each gate stack **606** may include an interfacial dielectric layer **612**, a gate dielectric layer **614** disposed on the interfacial dielectric layer **612**, and one or more conformal layers **616** disposed on the gate dielectric layer **614**. The gate electrode layer **610** may be disposed on the one or more conformal layers **616**. The interfacial dielectric layer **612** may include a dielectric material, such as an oxygen-containing material or a nitrogen-containing material, or multilayers thereof, and may be formed by any suitable deposition method, such as CVD, PECVD, or ALD. The gate dielectric layer **614** may include a dielectric material such as an oxygen-containing material or a nitrogen-containing material, a high-k dielectric material having a k value greater than that of silicon dioxide, or multilayers thereof. The gate dielectric layer **614** may be formed by any suitable method, such as CVD, PECVD, or ALD. The one or more conformal layers **616** may include one or more barrier layers and/or capping layers, such as a nitrogen-containing material, for example tantalum nitride (TaN), titanium nitride (TiN), or the like. The one or more conformal layers **616** may further include one or more work-function layers, such as aluminum titanium carbide, aluminum titanium oxide, aluminum titanium nitride, or the like. The term "conformal" may be used herein for ease of description upon a layer having substantial same thickness over various regions. The one or more conformal layers **616** may be deposited by ALD, PECVD, MBD, or any suitable deposition technique.

Gate spacers **618** are formed along sidewalls of the gate stacks **606** (e.g., sidewalls of the gate dielectric layers **614**). The gate spacers **618** may include silicon oxycarbide, silicon nitride, silicon oxynitride, silicon carbon nitride, the like, multi-layers thereof, or a combination thereof, and may be deposited by CVD, ALD, or other suitable deposition technique.

Portions of the gate stacks **606** and the gate spacers **618** may be formed on isolation regions **603**. The isolation regions **603** are formed on the substrate **601**. The isolation regions **603** may include an insulating material such as an oxygen-containing material, a nitrogen-containing material, or a combination thereof. The insulating material may be formed by a high-density plasma chemical vapor deposition (HDP-CVD), a flowable chemical vapor deposition (FCVD), or other suitable deposition process. In one aspect, the isolation regions **603** includes silicon oxide that is formed by a FCVD process.

A contact etch stop layer (CESL) **624** is formed on a portion of the S/D regions **604** and the isolation region **603**, and a first interlayer dielectric (ILD) **626** is formed on the CESL **624**. The CESL **624** can provide a mechanism to stop an etch process when forming openings in the first ILD **626**. The CESL **624** may be conformally deposited on surfaces of the S/D regions **604** and the isolation regions **603**. The CESL

624 may include an oxygen-containing material or a nitrogen-containing material, such as silicon nitride, silicon carbon nitride, silicon oxynitride, carbon nitride, silicon oxide, silicon carbon oxide, or the like, or a combination thereof, and may be deposited by CVD, PECVD, ALD, or any suitable deposition technique. The first ILD **626** may include an oxide formed by tetraethylorthosilicate (TEOS), undoped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), organosilicate glass (OSG), SiOC, and/or any suitable low-k dielectric materials (e.g., a material having a dielectric constant lower than silicon dioxide), and may be deposited by spin-on, CVD, FCVD, PECVD, PVD, or any suitable deposition technique.

A silicide layer **620** is formed on at least a portion of each S/D region **604**, as shown in FIGS. 2A and 2B. The silicide layer **620** may include a material having one or more of WSi, CoSi, NiSi, TiSi, MoSi and TaSi. In some embodiments, the silicide layer **620** includes a metal or metal alloy silicide, and the metal includes a noble metal, a refractory metal, a rare earth metal, alloys thereof, or combinations thereof. A conductive contact **622** is disposed on each silicide layer **620**. The conductive contact **622** may include a material having one or more of Ru, Mo, Co, Ni, W, Ti, Ta, Cu, Al, TiN or TaN, and the conductive contact **622** may be formed by any suitable method, such as electro-chemical plating (ECP), or PVD. The silicide layer **620** and the conductive contact **622** may be formed by first forming an opening in the first ILD **626** and the CESL **624** to expose at least a portion of the S/D region **604**, then forming the silicide layer **620** on the exposed portion of the S/D region **604**, and then forming the conductive contact **622** on the silicide layer **620**.

The dielectric material **104** may be formed over the devices **602**, as shown in FIGS. 2A and 2B. The conductive features **106** are formed in the dielectric material **104**, and each conductive feature **106** may be in contact with a corresponding conductive contact **622**.

FIGS. 3A-3M are cross-sectional side views of various stages of manufacturing an interconnect structure **101** along line A-A of FIG. 1, in accordance with some embodiments. The interconnect structure **101** may be formed over various devices of a semiconductor device structure **100** shown in FIGS. 1, 2A and 2B. For example, the interconnect structure **101** may be formed over one or more devices, such as transistors, diodes, imaging sensors, resistors, capacitors, inductors, memory cells, combinations thereof, and/or other suitable devices. In some embodiments, the interconnect structure **101** may be formed over the transistors, such as nanostructure FET having a plurality of channels wrapped around by a gate electrode layer.

As shown in FIG. 3A, the interconnection structure **101** includes the dielectric layer **102**, which may be an ILD layer or an intermetal dielectric (IMD) layer. The dielectric layer **102** includes the dielectric material **104**, the one or more first conductive features **106** (one conductive feature is shown) disposed in the dielectric material **104**, and an optional cap layer (not shown) disposed on each first conductive feature **106**. In some embodiments, the dielectric material **104** includes silicon oxide. The dielectric material **104** may be formed by chemical vapor deposition (CVD), atomic layer deposition (ALD), spin coating, or other suitable process. The first conductive feature **106** may each include an electrically conductive material, such as Cu, Co, Ru, Mo, Cr, W, Mn, Rh, Ir, Ni, Pd, Pt, Ag, Au, Al, alloys thereof, or other suitable material. In some embodiments, the first conductive feature **106** and the cap layer each includes a metal. The first

conductive feature **106** may be formed by physical vapor deposition (PVD), CVD, ALD, or other suitable process. The first conductive features **106** may be electrically connected to conductive contacts beneath the interconnection structure **101**.

As shown in FIG. 3B, a conductive structure **108** is formed over the dielectric material **104** and the first conductive features **106**. The conductive structure **108** may include a first conductive layer **110**, a second conductive layer **112** and a third conductive layer **114**. The first conductive layer **110** is a low electrical resistance material. In some embodiments, the first conductive layer **110** may include cobalt (Co), ruthenium (Ru), copper (Cu), chromium (Cr), titanium (Ti), vanadium (V), palladium (Pd), indium (In), alloys thereof, or other suitable material. In some embodiments, the first conductive layer **110** may be formed by PVD, ALD, or other suitable process under a process temperature between about 10° C. and 400° C. In some embodiments, the first conductive layer **110** may have a thickness ranging between about 10 Angstroms and 30 Angstroms.

The second conductive layer **112** is a low electrical resistance material. The second conductive layer **112** and the first conductive layer **110** include different materials. In some embodiments, the second conductive layer **112** may include tungsten (W), molybdenum (Mo), osmium (Os), iridium (Ir), cobalt (Co), niobium (Nb), platinum (Pt), rhodium (Rh), rhenium (Re), alloys thereof, or other suitable material. In some embodiments, the second conductive layer **112** may be formed by PVD, ALD, or other suitable process under a process temperature between about 10° C. and 400° C. In some embodiments, the second conductive layer **112** may have a thickness ranging between about 200 Angstroms and 500 Angstroms.

The third conductive layer **114** is a low electrical resistance material. The third conductive layer **114** and the second conductive layer **112** include different materials. In some embodiments, the third conductive layer **114** may include Ru, Cu, Cr, Ti, V, Pd, In, alloys thereof, or other suitable material. In some embodiments, the third conductive layer **114** may be formed by PVD, ALD, or other suitable process under a process temperature between about 10° C. and 400° C. In some embodiments, the third conductive layer **114** may have a thickness ranging between about 10 Angstroms and 80 Angstroms.

As shown in FIG. 3B, after forming the conductive structure **108**, a hard mask structure **116** is formed on the conductive structure **108**. The hard mask structure **116** includes a hard mask layer **118**, a bottom layer **120**, and a photoresist layer **122**. In some embodiments, the hard mask layer **118** may include SiO₂, SiCN, Si, SiC, SiN, SiON, TiN, Al, AlO_x, RuO₂, or other suitable material. In some embodiments, the bottom layer **120** may include a carbon-based material. In some embodiments, the bottom layer **120** may include a carbon-containing material that is easily removed by a plasma operation. In some embodiments, the photoresist layer **122** is used to pattern the bottom layer **120**. In some embodiments, the bottom layer **120** has a high etch selectivity to the hard mask layer **118**. The hard mask structure **116**, including the hard mask layer **118**, the bottom layer **120** and the photoresist layer **122**, allows the robust patterning of underlying layers.

As shown in FIG. 3C, a patterning operation is performed on the photoresist layer **122**. The photoresist layer **122** may be patterned using any suitable photolithography technique. For example, a photomask (not shown) may be disposed over the photoresist layer **122**, which may then be exposed

to a radiation beam including an ultraviolet (UV) or an excimer laser. In some embodiments, exposure of the photoresist layer 122 may be performed using an immersion lithography system to increase resolution and decrease the minimum achievable pitch. In some embodiments, a bake or cure operation may be performed to harden the photoresist layer 122, and a developer may be used to remove either the exposed or unexposed portions of the photoresist layer 122 depending on whether a positive or negative resist is used. In some embodiments, the hard mask structure 116 may further include anti-reflective materials (e.g., a backside anti-reflective coating (BARC) layer) to aid the exposure and focus of the photoresist layer 122.

As shown in FIG. 3D, the patterned photoresist layer 122 is used to pattern the bottom layer 120, and the pattern of the photoresist layer 122 is transferred to the bottom layer 120. The pattern may be transferred, for example, by one or more selective etching processes. In some embodiments, the bottom layer 120 may be patterned by a dry etch process. Because the bottom layer 120 includes a carbon-based material, the bottom layer 120 is easily patterned by a dry etch process. In some embodiments, the bottom layer 120 may be patterned by a dry etch process using Cl_2 , SiCl_4 , BCl_3 , N_2 , O_2 , H_2 , HBr , He , Ar , or other suitable gas. In some embodiments, the dry etch process to pattern the bottom layer 120 may use a transformer-coupled plasma (TCP) operation with a power between about 100 watts to 1800 watts. In some embodiments, the dry etch process to pattern the bottom layer 120 may use a TCP operation with a bias voltage under 300 volts.

As shown in FIG. 3E, after patterning the bottom layer 120, the patterned bottom layer 120 is used to pattern the hard mask layer 118. The hard mask layer 118 may be patterned by a dry etch, a wet etch, or other suitable etch process. In some embodiments, the hard mask layer 118 may be patterned by a dry etch process using CF_4 , CHF_3 , CH_3F , C_4F_8 , C_4F_6 , Cl_2 , SiCl_4 , BCl_3 , N_2 , O_2 , H_2 , HBr , He , Ar , or other suitable gas. In some embodiments, the hard mask layer 118 may be patterned by a TCP operation with a power between about 100 watts to 2000 watts. In some embodiments, the hard mask layer 118 may be patterned by a TCP operation with a bias voltage under 800 volts. In some embodiments, the hard mask layer 118 may be patterned by a wet etch process using DI , H_2O_2 , H_2O , HF , H_2PO_4 , HCl , CH_2CHOOH , H_2SO_4 , HNO_3 , HF , dHF , or other suitable material. In some embodiments, the bottom layer 120 has a high etch selectivity to the hard mask layer 118.

As shown in FIG. 3F, after patterning the hard mask layer 118, the photoresist layer 122 and the bottom layer 120 are removed. The patterned hard mask layer 118 is then used to pattern the conductive structure 108. In some embodiments, when patterning the bottom layer 120, the bottom layer 120 has a high etch selectivity to the hard mask layer 118. In some embodiments, when patterning the hard mask layer 118, the hard mask layer 118 has a high etch selectivity to the third conductive layer 114.

As shown in FIG. 3G, the hard mask layer 118 is then used to pattern the third conductive layer 114. The third conductive layer 114 may be patterned by a dry etch, a wet etch, or other suitable process. In some embodiments, the third conductive layer 114 may be patterned by a reactive-ion etching (RIE) process. In some embodiments, the third conductive layer 114 may be patterned by a dry etch process using chlorine based etching gas. In some embodiments, the third conductive layer 114 may be patterned by a dry etch process using Cl_2 , SiCl_4 , BCl_3 , N_2 , O_2 , H_2 , HBr , He , Ar , or other suitable gas. In some embodiments, the third conduc-

tive layer 114 may be patterned by a TCP operation with a power between about 100 watts to 1800 watts. In some embodiments, the third conductive layer 114 may be patterned by a TCP operation with a bias voltage under 300 volts. In some embodiments, when patterning the third conductive layer 114, the third conductive layer 114 has a high etch selectivity to the second conductive layer 112, and the third conductive layer 114 also has a high etch selectivity to the hard mask layer 118. In some embodiments, when patterning the third conductive layer 114, the third conductive layer 114 has an etch selectivity higher than 7 to the second conductive layer 112. In some embodiments, when patterning the third conductive layer 114, the third conductive layer 114 has an etch selectivity higher than 10 to the hard mask layer 118. As shown in FIG. 3G, after patterning the third conductive layer 114, a portion of the hard mask layer 118 is removed as well. Furthermore, because the third conductive layer 114 is thin and has a thickness ranging from about 10 Angstroms to about 80 Angstroms, over-etching of the third conductive layer 114 may be avoided. As a result, in combination with the etch selectivity between the second conductive layer 112 and the third conductive layer 114, the second conductive layer 112 may not be substantially affected by the patterning process of the third conductive layer 114.

As shown in FIG. 3H, the hard mask layer 118 is then used to pattern the second conductive layer 112. The second conductive layer 112 may be patterned by a dry etch, a wet etch, or other suitable process. In some embodiments, the second conductive layer 112 may be patterned by a RIE process. In some embodiments, the second conductive layer 112 may be patterned by a dry etch process using fluorine based etching gas. In some embodiments, the second conductive layer 112 may be patterned by a dry etch process using Cl_2 , SiCl_4 , BCl_3 , CF_4 , CHF_3 , CH_3F , CH_2F_2 , C_4F_8 , C_4F_6 , N_2 , O_2 , H_2 , HBr , He , Ar , or other suitable gas. In some embodiments, the second conductive layer 112 may be patterned by a TCP operation with a power between about 100 watts to 2000 watts. In some embodiments, the second conductive layer 112 may be patterned by a TCP operation with a bias voltage under 800 volts. In some embodiments, when patterning the second conductive layer 112, the second conductive layer 112 has a high etch selectivity to the first conductive layer 110, and the second conductive layer 112 also has a high etch selectivity to the hard mask layer 118. In some embodiments, when patterning the second conductive layer 112, the second conductive layer 112 has an etch selectivity higher than 8 to the first conductive layer 110. In some embodiments, when patterning the second conductive layer 112, the second conductive layer 112 has an etch selectivity higher than 5 to the hard mask layer 118. As shown in FIG. 3H, after patterning the second conductive layer 112, a further portion of the hard mask layer 118 is removed as well.

As shown in FIG. 3I, the hard mask layer 118 is then used to pattern the first conductive layer 110. The first conductive layer 110 may be patterned by a dry etch, a wet etch, or other suitable process. In some embodiments, the first conductive layer 110 may be patterned by a RIE process. In some embodiments, the first conductive layer 110 may be patterned by a dry etch process using Cl_2 , SiCl_4 , BCl_3 , N_2 , O_2 , H_2 , HBr , He , Ar , or other suitable gas. In some embodiments, the first conductive layer 110 may be patterned by a TCP operation with a power between about 100 watts to 1800 watts. In some embodiments, the first conductive layer 110 may be patterned by a TCP operation with a bias voltage under 300 volts. In some embodiments, when patterning the

first conductive layer 110, the first conductive layer 110 has a high etch selectivity to the dielectric material 104 and the first conductive feature 106, and the first conductive layer 110 also has a high etch selectivity to the hard mask layer 118. In some embodiments, when patterning the first conductive layer 110, the first conductive layer 110 has an etch selectivity higher than 10 to the dielectric material 104. In some embodiments, when patterning the first conductive layer 110, the first conductive layer 110 has an etch selectivity higher than 8 to the hard mask layer 118. As shown in FIG. 3I, after patterning the first conductive layer 110, a further portion of the hard mask layer 118 is removed as well. Furthermore, because the first conductive layer 110 is thin and has a thickness ranging from about 10 Angstroms to about 30 Angstroms, over-etching of the first conductive layer 110 may be avoided. As a result, in combination with the etch selectivity between the first conductive layer 110 and the first conductive feature 106, the first conductive feature 106 may not be substantially affected by the patterning process of the first conductive layer 110.

As shown in FIG. 3J, the hard mask layer 118 is removed, and a second conductive feature 130 and a third conductive feature 140 are formed on the layer 102. The second conductive feature 130 is formed over the first conductive feature 106, and third conductive feature 140 is formed over the dielectric material 104. The second conductive feature 130 includes a conductive layer 132, a conductive layer 134 and a conductive layer 136 stacking over the first conductive feature 106. The third conductive feature 140 includes a conductive layer 142, a conductive layer 144 and a conductive layer 146 stacking over the dielectric material 104. Since the second conductive feature 130 and the third conductive feature 140 are formed together from the conductive structure 108, the conductive layer 132 has the same material with the conductive layer 142, the conductive layer 134 has the same material with the conductive layer 144, and the conductive layer 136 has the same material with the conductive layer 146. Furthermore, the conductive layer 132, the conductive layer 134 and the conductive layer 136 have substantially the same width, and the conductive layer 142, the conductive layer 144 and the conductive layer 146 have substantially the same width.

Because the second conductive feature 130 and the third conductive feature 140 have similar structures, the second conductive feature 130 is used to explain the characteristics of the second conductive feature 130 and the third conductive feature 140 below.

As described above, in some embodiments, the first conductive layer 110 may have a thickness ranging between about 10 Angstroms to 30 Angstroms, the second conductive layer 112 may have a thickness ranging between about 200 Angstroms to 500 Angstroms, and the third conductive layer 114 may have a thickness ranging between about 10 Angstroms to 80 Angstroms. Hence, in some embodiments, the conductive layer 132 may have a thickness ranging between about 10 Angstroms to 30 Angstroms, the conductive layer 134 may have a thickness ranging between about 200 Angstroms to 500 Angstroms, and the conductive layer 136 may have a thickness ranging between about 10 Angstroms to 80 Angstroms.

By forming first conductive layer 110 under the second conductive layer 112, the first conductive layer 110 may perform as an etch stop layer when patterning the second conductive layer 112. Because the second conductive layer 112 has a high etch selectivity to the first conductive layer 110 when patterning the second conductive layer 112, and the first conductive layer 110 has a high etch selectivity to

the dielectric material 104 and the first conductive feature 106 when patterning the first conductive layer 110, the first conductive layer 110 may prevent the damage to the dielectric material 104 and the first conductive feature 106 when forming the second conductive feature 130 and the third conductive feature 140. Furthermore, the via damage of the first conductive feature 106 caused by the oxide recess of the dielectric material 104 may be also prevented.

If the conductive layer 132 (the first conductive layer 110) is too thin, such as less than about 10 Angstroms, the conductive layer 132 may not provide sufficient protection to the dielectric material 104 and the first conductive feature 106 when forming the second conductive feature 130 and the third conductive feature 140. If the conductive layer 132 is too thick, such as greater than about 30 Angstroms, the patterning process forming the conductive layer 132 may also damage the first conductive feature 106, and the electrical properties (e.g., conductivity) of the second conductive feature 130 and the third conductive feature 140 may be worsened. Therefore, in some embodiments, the conductive layer 132 may have a thickness ranging between about 10 Angstroms to 30 Angstroms, and a thickness of the conductive layer 132 is about 3% to 8% of a total thickness of the second conductive feature 130. Similarly, if the thickness of the conductive layer 132 is less than about 3% of the total thickness of the second conductive feature 130, the conductive layer 132 may not provide sufficient protection to the dielectric material 104 and the first conductive feature 106. On the other hand, if the thickness of the conductive layer 132 is greater than about 8% of the total thickness of the second conductive feature 130, the patterning process forming the conductive layer 132 may also damage the first conductive feature 106, and the electrical properties (e.g., conductivity) of the second conductive feature 130 and the third conductive feature 140 may be worsened.

Because the etch selectivity of second conductive layer 112 to the hard mask layer 118 is insufficient, in a situation that patterning the second conductive layer 112 without the third conductive layer 114, the second conductive layer 112 may be damaged and the metal lines formed by the second conductive feature 130 and the third conductive feature 140 may be broken. By forming the third conductive layer 114 above the second conductive layer 112, the third conductive layer 114 may provide a protection to the second conductive layer 112 when patterning the second conductive layer 112.

If the conductive layer 136 (the third conductive layer 114) is too thin, the conductive layer 136 may not provide sufficient protection to the conductive layer 134 (the second conductive layer 112) when forming the second conductive feature 130 and the third conductive feature 140. If the conductive layer 136 is too thick, the patterning process forming the conductive layer 136 may also damage the conductive layer 134, and the electrical properties (e.g., conductivity) of the second conductive feature 130 and the third conductive feature 140 may be worsened. Therefore, in some embodiments, the conductive layer 136 may have a thickness ranging between about 10 Angstroms to 80 Angstroms, and a thickness of the conductive layer 136 is about 5% to 15% of a total thickness of the second conductive feature 130. Similarly, if the thickness of the conductive layer 136 is less than about 5% of the total thickness of the second conductive feature 130, the conductive layer 136 may not provide sufficient protection to the conductive layer 134. On the other hand, if the thickness of the conductive layer 136 is greater than about 15% of the total thickness of the second conductive feature 130, the patterning process forming the conductive layer 136 may also damage the

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conductive layer **134**, and the electrical properties (e.g., conductivity) of the second conductive feature **130** and the third conductive feature **140** may be worsened.

As shown in FIG. 3K, a dielectric fill **150** is disposed over the dielectric layer **102**, the second conductive feature **130** and the third conductive feature **140**.

The dielectric fill **150** may be a silicon-containing material, such as SiCO, SiCN, SiN, SiCON, SiO_x, SiC, SiCOH, or SiON. In some embodiments, the dielectric fill **150** includes a low-k dielectric material having a k value ranging from about 2 to about 3.6, such as SiCOH. The low-k dielectric material may have a porosity ranging from about 0.1 percent to about 40 percent. The dielectric fill **150** may fill the space over the dielectric layer **102** between the second conductive feature **130** and the third conductive feature **140**. The dielectric fill **150** may also cover the second conductive feature **130** and the third conductive feature **140**. The dielectric fill **150** may be formed by CVD, ALD, PECVD, PEALD, or other suitable process.

In some embodiments, before forming the dielectric fill **150**, a capping layer (not shown) may be formed on the exposed surfaces of the dielectric material **104**, the second conductive feature **130** and the third conductive feature **140**. The capping layer may prevent metal diffusion from the second conductive feature **130** and the third conductive feature **140** to the dielectric fill **150**. The capping layer may be made of a dielectric material. In some embodiments, the capping layer includes SiCO, SiCN, SiN, SiCON, SiO_x, SiC, SiON, or other suitable dielectric materials. The capping layer may be formed by any suitable process, such as PVD, ALD, CVD, PECVD, PEALD or any suitable conformal process. The capping layer may have a thickness ranging from about 5 Angstroms to about 200 Angstroms.

As shown in FIG. 3L, a planarization process may be performed to remove a portion of the dielectric fill **150**. In some embodiments, the planarization process may also remove a portion of the conductive layer **136** and a portion of the conductive layer **146**. The planarization process may be any suitable process, such as a chemical-mechanical polishing (CMP) process. As a result of the planarization process, a top surface of the dielectric fill **150**, the second conductive feature **130** and the third conductive feature **140** may be substantially co-planar.

Then, as shown in FIG. 3M, an etch stop layer **160** may be formed on the dielectric fill **150**, the second conductive feature **130** and the third conductive feature **140**. The etch stop layer **160** is formed on the planar surfaces of the dielectric fill **150**, the second conductive feature **130** and the third conductive feature **140**. The etch stop layer **160** may be a single layer or a multi-layer structure. The etch stop layer **160** may include a metal oxide, such as Al, Zr, Y, Hf, or other suitable metal oxide, or a silicon-containing material, such as SiCO, SiCN, SiN, SiCON, SiO_x, SiC, SiON, or the like. The etch stop layer **160** may be formed by PVD, CVD, ALD, spin-on, or any suitable deposition process. The etch stop layer **160** may have a thickness ranging from about 1 Angstrom to about 100 Angstroms.

FIGS. 4A-4B are cross-sectional side views of various stages of manufacturing an interconnect structure **200**, in accordance with some embodiments. The interconnect structure **200** is similar to the interconnect structure **101** in FIGS. 3A-3M. However, as shown in FIG. 4A, the second conductive feature **130** may include the conductive layer **132** and the conductive layer **134**, and the third conductive feature **140** may include the conductive layer **142** and the conductive layer **144**. In other words, the second conductive

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feature **130** and the third conductive feature **140** in the interconnect structure **200** are two-layer structures.

In some embodiments, in the planarization process shown in FIG. 3L, the planarization process may remove a portion of the dielectric fill **150**, and further remove all the conductive layer **136** and all the conductive layer **146** to expose the conductive layer **134** and the conductive layer **144**. As a result of the planarization process, a top surface of the dielectric fill **150**, the conductive layer **134** and the conductive layer **144** may be substantially co-planar.

Then, as shown in FIG. 4B, the etch stop layer **160** may be formed on the dielectric fill **150**, the second conductive feature **130** and the third conductive feature **140**. The etch stop layer **160** is formed on the planar surfaces of the dielectric fill **150**, the conductive layer **134** and the conductive layer **144**.

Because the function of the conductive layer **136** and the conductive layer **146** (the third conductive layer **114** in FIGS. 3B-3I) is to provide a protection to the second conductive layer **112** when patterning the second conductive layer **112**. During the planarization process shown in FIG. 3L, the patterning process of the second conductive layer **112** is completed, and therefore the conductive layer **136** and the conductive layer **146** may be optionally removed.

In another embodiment, when the design of the interconnect structure **200** provides a protection to prevent the damage to the dielectric material **104** and the first conductive feature **106** when forming the second conductive feature **130** and the third conductive feature **140**, the conductive structure **108** may be a two-layer structure including the first conductive layer **110** and the second conductive layer **112**. In this situation, after forming the second conductive feature **130** and the third conductive feature **140**, the second conductive feature **130** and the third conductive feature **140** in the interconnect structure **200** are two-layer structures, and the third conductive layer **114** (the conductive layers **136**, **146**) is not present.

FIGS. 5A-5B are cross-sectional side views of various stages of manufacturing an interconnect structure **300**, in accordance with some embodiments. The interconnect structure **300** is similar to the interconnect structure **101** in FIGS. 3A-3M. However, as shown in FIG. 5A, the second conductive feature **130** may include the conductive layer **134** and the conductive layer **136**, and the third conductive feature **140** may include the conductive layer **144** and the conductive layer **146**. In other words, the second conductive feature **130** and the third conductive feature **140** in the interconnect structure **300** are two-layer structures, and the conductive layers **132**, **142** are not present.

In some embodiments, when the design of the interconnect structure **300** provides a protection to the second conductive layer **112** when patterning the first conductive layer **110** to prevent the damage to the metal lines formed by the second conductive feature **130** and the third conductive feature **140**, the conductive structure **108** may be a two-layer structure including the second conductive layer **112** and the third conductive layer **114**. Then, after the planarization process, as shown in FIG. 5A, a top surface of the dielectric fill **150**, the conductive layer **136** and the conductive layer **146** may be substantially co-planar.

Then, as shown in FIG. 5B, the etch stop layer **160** may be formed on the dielectric fill **150**, the second conductive feature **130** and the third conductive feature **140**. The etch stop layer **160** is formed on the planar surfaces of the dielectric fill **150**, the conductive layer **136** and the conductive layer **146**.

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FIG. 6 is a flow chart of a method 400 of manufacturing the interconnect structure 200 or the interconnect structure 300 in accordance with some embodiments. It is noted that the operations of the method 400, including any descriptions given with reference to the figures, are merely exemplary and are not intended to be limiting beyond what is specifically recited in the claims that follow. Additional operations may be implemented before, during, and after the method 400, and some operations may be replaced, eliminated, or rearranged in any desired order in accordance with various embodiments of the method 400.

The method 400 starts at operation 402 by forming a first conductive feature in a dielectric layer. The first conductive feature may be the conductive feature 106 and the dielectric layer may be the dielectric material 104. The first conductive feature and the dielectric layer may be formed by the processes discussed above with respect to FIG. 3A.

At operation 404, a first conductive layer, a second conductive layer and a hard mask structure are sequentially formed over the first conductive feature and the dielectric layer. The first conductive layer and the second conductive layer may be formed by PVD, ALD, or other suitable process.

When forming the interconnect structure 200, the first conductive layer and the second conductive layer may be the first conductive layer 110 and the second conductive layer 112. When forming the interconnect structure 200, the first conductive layer may be formed by Co, Ru, Cu, Cr, Ti, V, Pd, In, alloys thereof, or other suitable material. When forming the interconnect structure 200, the second conductive layer may be formed by W, Mo, Os, Jr, Co, Nb, Pt, Rh, Re, alloys thereof, or other suitable material.

When forming the interconnect structure 300, the first conductive layer and the second conductive layer may be the second conductive layer 112 and the third conductive layer 114. When forming the interconnect structure 300, the first conductive layer may be formed by W, Mo, Os, Ir, Co, Nb, Pt, Rh, Re, alloys thereof, or other suitable material. When forming the interconnect structure 300, the second conductive layer may be formed by Ru, Cu, Cr, Ti, V, Pd, In, alloys thereof, or other suitable material.

The hard mask structure may be the hard mask structure 116. The hard mask structure may include a hard mask layer, a bottom layer, and a photoresist layer. In some embodiments, the hard mask layer, the bottom layer, and the photoresist layer may be the hard mask layer 118, the bottom layer 120, and the photoresist layer 122.

In some embodiments, the hard mask layer may include SiO₂, SiCN, Si, SiC, SiN, SiON, TiN, Al, AlO_x, RuO₂, or other suitable material. In some embodiments, the bottom layer may include a carbon-based material. In some embodiments, the bottom layer may include a carbon-containing material that is easily removed by a plasma operation. In some embodiments, the photoresist layer is used to pattern the bottom layer. In some embodiments, the bottom layer has a high etch selectivity to the hard mask layer. The hard mask structure, including the hard mask layer, the bottom layer, and the photoresist layer, allows the robust patterning of underlying layers.

At operation 406, a portion of the hard mask structure is removed to expose a portion of the second conductive layer. The removal of the portion of the hard mask structure may include multiple operations.

A patterning operation is first performed on the photoresist layer. The photoresist layer may be patterned using any suitable photolithography technique. In some embodiments, a bake or cure operation may be performed to harden the

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photoresist layer, and a developer may be used to remove either the exposed or unexposed portions of the photoresist layer depending on whether a positive or negative resist is used. In some embodiments, the hard mask structure may further include anti-reflective materials (e.g., a BARC layer) to aid the exposure and focus of the photoresist layer. Then, the photoresist layer is used to pattern the bottom layer, and the pattern of the photoresist layer is transferred to the bottom layer. The pattern may be transferred, for example, by one or more selective etching processes. After patterning the bottom layer, the bottom layer is used to pattern the hard mask layer. The hard mask layer may be patterned by a dry etch, a wet etch, or other suitable etch process. In some embodiments, the bottom layer has a high etch selectivity to the hard mask layer. After patterning the hard mask layer, the photoresist layer and the bottom layer are removed.

At operation 408, a first etch process is performed to remove the exposed portion of the second conductive layer and expose a portion of the first conductive layer. The first etch process may be a dry etch, a wet etch, or other suitable process. The first etch process may be a RIE process.

When forming the interconnect structure 200, the first etch process may be a dry etch process using Cl₂, SiCl₄, BCl₃, CF₄, CHF₃, CH₃F, CH₂F₂, C₄F₈, C₄F₆, N₂, O₂, H₂, HBr, He, Ar, or other suitable gas. When forming the interconnect structure 200, the first etch process may be a TCP operation with a power between about 100 watts to 2000 watts. When forming the interconnect structure 200, the first etch process may be a TCP operation with a bias voltage under 800 volts.

When forming the interconnect structure 300, the first etch process may be a dry etch process using Cl₂, SiCl₄, BCl₃, N₂, O₂, H₂, HBr, He, Ar, or other suitable gas. When forming the interconnect structure 300, the first etch process may be a TCP operation with a power between about 100 watts to 1800 watts. When forming the interconnect structure 300, the first etch process may be a TCP operation with a bias voltage under 300 volts.

At operation 410, a second etch process is performed to remove the exposed portion of the first conductive layer and expose a portion of the dielectric layer. The second etch process may be a dry etch, a wet etch, or other suitable process. The second etch process may be a RIE process.

When forming the interconnect structure 200, the second etch process may be a dry etch process using Cl₂, SiCl₄, BCl₃, N₂, O₂, H₂, HBr, He, Ar, or other suitable gas. When forming the interconnect structure 200, the first etch process may be a TCP operation with a power between about 100 watts to 1800 watts. When forming the interconnect structure 200, the first etch process may be a TCP operation with a bias voltage under 300 volts.

When forming the interconnect structure 300, the second etch process may be a dry etch process using Cl₂, SiCl₄, BCl₃, CF₄, CHF₃, CH₃F, CH₂F₂, C₄F₈, C₄F₆, N₂, O₂, H₂, HBr, He, Ar, or other suitable gas. When forming the interconnect structure 300, the second etch process may be a TCP operation with a power between about 100 watts to 2000 watts. When forming the interconnect structure 300, the second etch process may be a TCP operation with a bias voltage under 800 volts.

In some embodiments, the second conductive layer and the first conductive layer have different etch selectivity in the first etch process. In some embodiments, the method 400 may further include removing the hard mask structure. Further, a dielectric fill may be formed over the exposed dielectric layer, the first conductive layer and the second conductive layer. Then, a planarization process may be

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performed to remove a portion of the dielectric fill, and a surface of the dielectric fill is substantially co-planar with a surface of the second conductive layer. Then, an etch stop layer may be formed over the surface of the dielectric fill and the surface of the second conductive layer.

FIG. 7 is another flow chart of a method 500 of manufacturing the interconnect structure 101 in accordance with some embodiments. It is noted that the operations of the method 500, including any descriptions given with reference to the figures, are merely exemplary and are not intended to be limiting beyond what is specifically recited in the claims that follow. Additional operations may be implemented before, during, and after the method 500, and some operations may be replaced, eliminated, or rearranged in any desired order in accordance with various embodiments of the method 500.

The method 500 starts at operation 502 by forming a first conductive feature in a dielectric layer. The first conductive feature may be the conductive feature 106 and the dielectric layer may be the dielectric material 104. The first conductive feature and the dielectric layer may be formed by the processes discussed above with respect to FIG. 3A.

At operation 504, a first conductive layer, a second conductive layer, a third conductive layer and a hard mask structure are sequentially formed over the first conductive feature and the dielectric layer. The first conductive layer, the second conductive layer and the third conductive layer may be formed by PVD, ALD, or other suitable process.

In some embodiments, the first conductive layer may be the first conductive layer 110, the second conductive layer may be the second conductive layer 112, and the third conductive layer may be the third conductive layer 114. In some embodiments, the first conductive layer may be formed by Co, Ru, Cu, Cr, Ti, V, Pd, In, alloys thereof, or other suitable material. In some embodiments, the second conductive layer may be formed by W, Mo, Os, Ir, Co, Nb, Pt, Rh, Re, alloys thereof, or other suitable material. In some embodiments, the third conductive layer may be formed by Ru, Cu, Cr, Ti, V, Pd, In, alloys thereof, or other suitable material.

The hard mask structure may be the hard mask structure 116. The hard mask structure may include a hard mask layer, a bottom layer, and a photoresist layer. In some embodiments, the hard mask layer, the bottom layer, and the photoresist layer may be the hard mask layer 118, the bottom layer 120, and the photoresist layer 122.

In some embodiments, the hard mask layer may include SiO₂, SiCN, Si, SiC, SiN, SiON, TiN, Al, AlO_x, RuO₂, or other suitable material. In some embodiments, the bottom layer may include a carbon-based material. In some embodiments, the bottom layer may include a carbon-containing material that is easily removed by a plasma operation. In some embodiments, the photoresist layer is used to pattern the bottom layer. In some embodiments, the bottom layer has a high etch selectivity to the hard mask layer. The hard mask structure, including the hard mask layer, the bottom layer, and the photoresist layer, allows the robust patterning of underlying layers.

At operation 506, a portion of the hard mask structure is removed to expose a portion of the third conductive layer. The removal of the portion of the hard mask structure may include multiple operations.

A patterning operation is first performed on the photoresist layer. The photoresist layer may be patterned using any suitable photolithography technique. In some embodiments, a bake or cure operation may be performed to harden the photoresist layer, and a developer may be used to remove

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either the exposed or unexposed portions of the photoresist layer depending on whether a positive or negative resist is used. In some embodiments, the hard mask structure may further include anti-reflective materials (e.g., a BARC layer) to aid the exposure and focus of the photoresist layer. Then, the photoresist layer is used to pattern the bottom layer, and the pattern of the photoresist layer is transferred to the bottom layer. The pattern may be transferred, for example, by one or more selective etching processes. After patterning the bottom layer, the bottom layer is used to pattern the hard mask layer. The hard mask layer may be patterned by a dry etch, a wet etch, or other suitable etch process. In some embodiments, the bottom layer has a high etch selectivity to the hard mask layer. After patterning the hard mask layer, the photoresist layer and the bottom layer are removed.

At operation 508, a first etch process is performed to remove the exposed portion of the third conductive layer and expose a portion of the second conductive layer. The first etch process may be a dry etch, a wet etch, or other suitable process. The first etch process may be a RIE process.

In some embodiments, the first etch process may be a dry etch process using Cl₂, SiCl₄, BCl₃, N₂, O₂, H₂, HBr, He, Ar, or other suitable gas. In some embodiments, the first etch process may be a TCP operation with a power between about 100 watts to 1800 watts. In some embodiments, the first etch process may be a TCP operation with a bias voltage under 300 volts.

At operation 510, a second etch process is performed to remove the exposed portion of the second conductive layer and expose a portion of the first conductive layer. The second etch process may be a dry etch, a wet etch, or other suitable process. The second etch process may be a RIE process. In some embodiments, the second etch process may be a dry etch process using Cl₂, SiCl₄, BCl₃, CF₄, CHF₃, CH₃F, CH₂F₂, C₄F₈, C₄F₆, N₂, O₂, H₂, HBr, He, Ar, or other suitable gas. In some embodiments, the second etch process may be a TCP operation with a power between about 100 watts to 2000 watts. In some embodiments, the second etch process may be a TCP operation with a bias voltage under 800 volts.

At operation 512, a third etch process is performed to remove the exposed portion of the first conductive layer and expose a portion of the dielectric layer. The third etch process may be a dry etch, a wet etch, or other suitable process. The third etch process may be a RIE process. In some embodiments, the third etch process may be a dry etch process using Cl₂, SiCl₄, BCl₃, N₂, O₂, H₂, HBr, He, Ar, or other suitable gas. In some embodiments, the third etch process may be a TCP operation with a power between about 100 watts to 1800 watts. In some embodiments, the third etch process may be a TCP operation with a bias voltage under 300 volts.

In some embodiments, the third conductive layer and the second conductive layer have different etch selectivity in the first etch process, and the second conductive layer and the first conductive layer have different etch selectivity in the second etch process.

In some embodiments, the method 500 may further include removing the hard mask structure. A dielectric fill may be formed over the exposed dielectric layer, the first conductive layer, the second conductive layer and the third conductive layer. Then, a planarization process may be performed to remove a portion of the dielectric fill, and a surface of the dielectric fill is substantially co-planar with a surface of the third conductive layer. Then, an etch stop layer may be formed over the surface of the dielectric fill and the surface of the third conductive layer.

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By forming first conductive layer under the second conductive layer, the first conductive layer may perform as an etch stop layer when patterning the second conductive layer. Because the second conductive layer has a high etch selectivity to the first conductive layer when patterning the second conductive layer, and the first conductive layer has a high etch selectivity to the dielectric layer when patterning the first conductive layer, the first conductive layer may prevent the damage to the dielectric layer when forming the second conductive feature and the third conductive feature. Further, the via damage of the first conductive feature caused by the oxide recess of the dielectric layer may be also prevented.

By forming the third conductive layer above the second conductive layer, the third conductive layer may provide a protection to the second conductive layer when patterning the second conductive layer to prevent the metal line damage.

An embodiment is an interconnect structure. The interconnect structure includes a dielectric layer, a first conductive feature, a second conductive feature, a third conductive feature, and a dielectric fill. The first conductive feature is disposed in the dielectric layer. The second conductive feature is disposed over the first conductive feature. The second conductive feature includes a first conductive layer disposed over the first conductive feature, a second conductive layer disposed on the first conductive layer, and a third conductive layer disposed on the second conductive layer. The first conductive layer, the second conductive layer and the third conductive layer have substantially the same width. The third conductive feature is disposed over the dielectric layer. The third conductive feature includes a fourth conductive layer disposed over the dielectric layer, a fifth conductive layer disposed on the fourth conductive layer, and a sixth conductive layer disposed on the fifth conductive layer. The fourth conductive layer, the fifth conductive layer and the sixth conductive layer have substantially the same width. The dielectric fill is disposed over the dielectric layer between the second conductive feature and the third conductive feature.

Another embodiment is a method for manufacturing an interconnection structure. A first conductive feature is formed in a dielectric layer. A first conductive layer, a second conductive layer and a hard mask structure are sequentially formed over the first conductive feature and the dielectric layer. A portion of the hard mask structure is removed to expose a portion of the second conductive layer. A first etch process is performed to remove the exposed portion of the second conductive layer and expose a portion of the first conductive layer. A second etch process is performed to remove the exposed portion of the first conductive layer and expose a portion of the dielectric layer.

A further embodiment is a method for manufacturing an interconnection structure. A first conductive feature is formed in a dielectric layer. A first conductive layer, a second conductive layer, a third conductive layer and a hard mask structure are sequentially formed over the first conductive feature and the dielectric layer. A portion of the hard mask structure is removed to expose a portion of the third conductive layer. A first etch process is performed to remove the exposed portion of the third conductive layer and expose a portion of the second conductive layer. A second etch process is performed to remove the exposed portion of the second conductive layer and expose a portion of the first conductive layer. A third etch process is performed to remove the exposed portion of the first conductive layer and expose a portion of the dielectric layer.

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The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

The invention claimed is:

1. An interconnection structure, comprising:

a dielectric layer;

a first conductive feature disposed in the dielectric layer;

a second conductive feature disposed over the first conductive feature, the second conductive feature comprising a first conductive layer disposed on and in contact with the first conductive feature, a second conductive layer disposed on and in contact with the first conductive layer, and a third conductive layer disposed on and in contact with the second conductive layer, wherein the first conductive layer, the second conductive layer and the third conductive layer have substantially the same width, and a thickness of the third conductive layer is about 5% to 15% of a total thickness of the second conductive feature;

a third conductive feature disposed over the dielectric layer, the third conductive feature comprising a fourth conductive layer disposed over the dielectric layer, a fifth conductive layer disposed on the fourth conductive layer, and a sixth conductive layer disposed on the fifth conductive layer, wherein the fourth conductive layer, the fifth conductive layer and the sixth conductive layer have substantially the same width;

a dielectric fill disposed over the dielectric layer between the second conductive feature and the third conductive feature; and

a continuous etch stop layer disposed on and in contact with the third conductive layer of the second conductive feature, the third conductive feature, and the dielectric fill.

2. The interconnection structure of claim 1, wherein the first conductive layer comprises cobalt (Co), ruthenium (Ru), copper (Cu), chromium (Cr), titanium (Ti), vanadium (V), palladium (Pd), indium (In) or an alloy thereof.

3. The interconnection structure of claim 2, wherein the first conductive layer has a thickness ranging from about 10 Angstroms to 30 Angstroms.

4. The interconnection structure of claim 1, wherein the first conductive layer comprises the same material as the fourth conductive layer.

5. The interconnection structure of claim 1, wherein the second conductive layer comprises tungsten (W), molybdenum (Mo), osmium (Os), iridium (Ir), cobalt (Co), niobium (Nb), platinum (Pt), rhodium (Rh), rhenium (Re) or an alloy thereof, wherein the second conductive layer and the first conductive layer comprise different materials.

6. The interconnection structure of claim 1, wherein the third conductive layer comprises ruthenium (Ru), copper (Cu), chromium (Cr), titanium (Ti), vanadium (V), palladium (Pd), indium (In) or an alloy thereof, wherein the second conductive layer and the third conductive layer comprise different materials.

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7. The interconnection structure of claim 6, wherein the third conductive layer has a thickness ranging from about 10 Angstroms to 80 Angstroms.

8. The interconnection structure of claim 1, wherein a thickness of the first conductive layer is about 3% to 8% of a total thickness of the second conductive feature. 5

9. The interconnection structure of claim 1, wherein the etch stop layer is disposed on and in contact with the sixth conductive layer of. 10

10. An interconnection structure, comprising:

a dielectric layer;

a first conductive feature disposed in the dielectric layer;

a second conductive feature disposed over the first conductive feature, the second conductive feature comprising a first conductive layer disposed on and in contact with the first conductive feature and a second conductive layer disposed on and in contact with the first conductive layer, wherein a thickness of the first conductive layer is about 3% to 8% of a total thickness of the second conductive feature; 15 20

a dielectric fill disposed over the dielectric layer and adjacent the second conductive feature, wherein the dielectric fill is in contact with the first and second conductive layers, and a top surface of the dielectric fill and a top surface of the second conductive layer are substantially co-planar; and 25

an etch stop layer disposed on the dielectric fill and the second conductive feature. 30

11. The interconnection structure of claim 10, wherein the first and second conductive layers comprise different metals.

12. The interconnection structure of claim 11, wherein the second conductive layer has an etch selectivity higher than 8 to the first conductive layer.

13. The interconnection structure of claim 11, wherein the first conductive layer has a thickness ranging from about 10 Angstroms to about 30 Angstroms. 35

14. The interconnection structure of claim 13, wherein the second conductive layer has a thickness ranging from about 200 Angstroms to about 500 Angstroms.

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15. An interconnection structure, comprising:

a first conductive feature disposed over a second conductive feature, wherein the first conductive feature comprises a first conductive layer disposed on and in contact with the second conductive feature, the first conductive layer comprises a first material, a second conductive layer disposed on and in contact with the first conductive layer, the second conductive layer comprises a second material different from the first material, and a third conductive layer disposed on and in contact with the second conductive layer, the third conductive layer comprises a third material different from the second material, wherein the first conductive layer, the second conductive layer and the third conductive layer have substantially the same width, and a thickness of the first conductive layer is about 3% to 8% of a total thickness of the first conductive feature; a third conductive feature disposed adjacent the first conductive feature, wherein the third conductive feature comprises a fourth conductive layer disposed adjacent the first conductive layer, a fifth conductive layer disposed on the fourth conductive layer, and a sixth conductive layer disposed on the fifth conductive layer, wherein the fourth conductive layer, the fifth conductive layer and the sixth conductive layer have substantially the same width; 15 20 25

a dielectric fill disposed between the first conductive feature and the third conductive feature; and

a continuous etch stop layer disposed on and in contact with the third conductive layer of the first conductive feature, the third conductive feature, and the dielectric fill. 30

16. The interconnection structure of claim 15, wherein top surfaces of the dielectric fill, the first conductive feature, and the third conductive feature are substantially co-planar.

17. The interconnection structure of claim 16, wherein the etch stop layer is disposed on and in contact with the sixth conductive layer of the third conductive feature. 35

18. The interconnection structure of claim 15, wherein a thickness of the third conductive layer is about 5% to 15% of a total thickness of the first conductive feature.

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